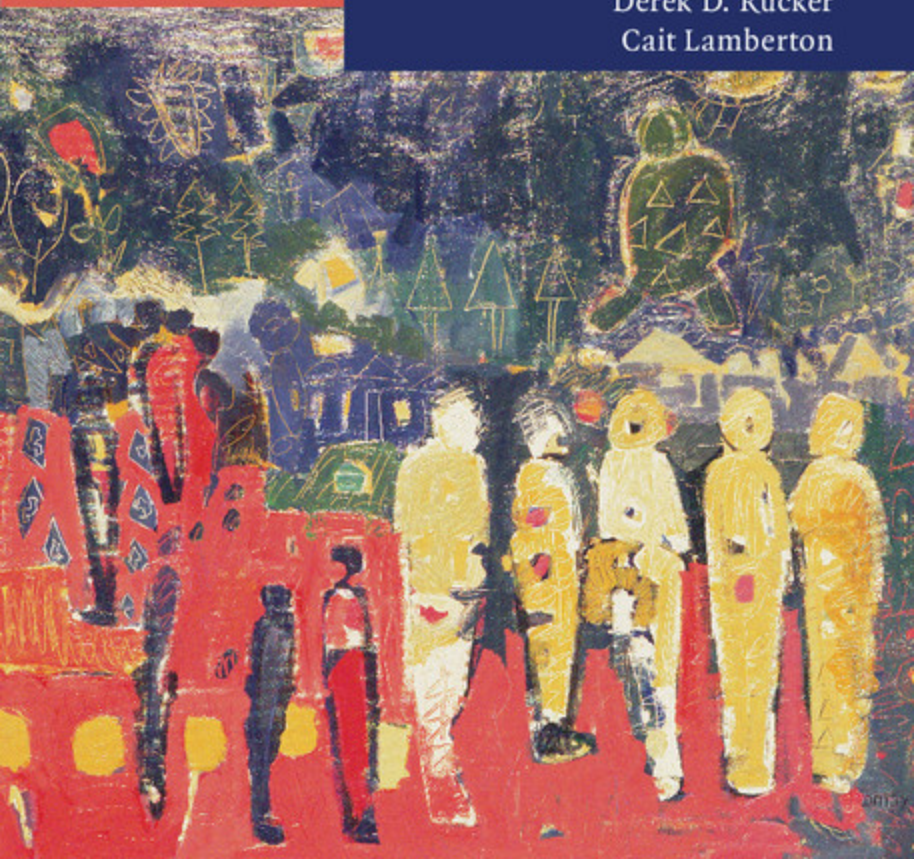


THE CAMBRIDGE HANDBOOK OF
**CONSUMER
PSYCHOLOGY**

Edited by Michael I. Norton
Derek D. Rucker
Cait Lambertson



The Cambridge Handbook of Consumer Psychology

Why do consumers make the purchases they do, and which ones make them truly happy? Why are consumers willing to spend huge sums of money to appear high status? This handbook addresses these key questions and many more. It provides a comprehensive overview of consumer psychology, examining cutting-edge research at the individual, interpersonal, and societal levels. Leading scholars summarize past and current findings and consider future lines of inquiry to deepen our understanding of the psychology behind consumers' decision making, their interactions with other consumers, and the effects of societal factors on consumption. *The Cambridge Handbook of Consumer Psychology* will act as a valuable guide for faculty as well as graduate and undergraduate students in psychology, marketing, management, sociology, and anthropology.

MICHAEL I. NORTON is a Professor of Business Administration in the Marketing Unit at Harvard Business School. He is the coauthor, with Elizabeth Dunn, of *Happy Money: The Science of Smarter Spending* (2013).

DEREK D. RUCKER is the Sandy and Morton Goldman Professor of Entrepreneurial Studies in Marketing at Northwestern University's Kellogg School of Management. His articles have appeared in numerous publications, including the *Journal of Personality and Social Psychology*, the *Journal of Consumer Research*, the *Journal of Marketing Research*, and the *Journal of Consumer Psychology*.

CAIT LAMBERTON is an Associate Professor and Fryrear Faculty Fellow in Marketing at the Katz Graduate School of Business at the University of Pittsburgh. Her work has been published in the *Journal of Consumer Research*, the *Journal of Consumer Psychology*, the *Journal of Marketing Research*, the *Journal of Marketing*, and the *Journal of Public Policy and Marketing*.

The Cambridge Handbook of Consumer Psychology

Edited by

Michael I. Norton

Derek D. Rucker

Cait Lamberton



CAMBRIDGE
UNIVERSITY PRESS

CAMBRIDGE
UNIVERSITY PRESS

32 Avenue of the Americas, New York, NY 10013-2473, USA

Cambridge University Press is part of the University of Cambridge.

It furthers the University's mission by disseminating knowledge in the pursuit of education, learning, and research at the highest international levels of excellence.

www.cambridge.org

Information on this title: www.cambridge.org/9781107069206

© Cambridge University Press 2015

This publication is in copyright. Subject to statutory exception and to the provisions of relevant collective licensing agreements, no reproduction of any part may take place without the written permission of Cambridge University Press.

First published 2015

Printed in the United States of America

A catalog record for this publication is available from the British Library.

Library of Congress Cataloging in Publication Data

The Cambridge handbook of consumer psychology / [edited by] Michael I. Norton, Harvard Business School, Harvard University, Derek D. Rucker, Kellogg School of Management, Northwestern University, Cait Lambertson, Katz Graduate School of Business, University of Pittsburgh.

pages cm. – (Cambridge handbooks in psychology)

Includes bibliographical references and index.

ISBN 978-1-107-06920-6 (Hardback)

1. Consumers–Psychology. 2. Consumer behavior. I. Norton, Michael I., 1975– editor. II. Rucker, Derek D., 1977– editor. III. Lambertson, Cait, 1975– editor. HF5415.32.C36 2015

658.8'342–dc23 2015006129

ISBN 978-1-107-06920-6 Hardback

Cambridge University Press has no responsibility for the persistence or accuracy of URLs for external or third-party Internet Web sites referred to in this publication and does not guarantee that any content on such Web sites is, or will remain, accurate or appropriate.

Contents

<i>List of Figures</i>	<i>page</i> viii
<i>List of Tables</i>	ix
<i>List of Contributors</i>	xi

Introduction: Understanding Consumers in the Here, the Now, and the Tomorrow MICHAEL I. NORTON, DEREK D. RUCKER, AND CAIT LAMBERTON	1
--	---

Part I. Individual Consumer Decision Making and Behavior

1 Consumer Happiness and Well-Being CASSIE MOGILNER AND MICHAEL I. NORTON	5
2 Attitude Change and Persuasion: Past, Present, and Future Directions ZAKARY L. TORMALA AND PABLO BRIÑOL	29
3 Consumer Prediction: Forecasted Utility, Psychological Distance, and Their Intersection CAREY K. MOREWEDGE AND HAL E. HERSHFELD	65
4 Consumer Emotions EDUARDO B. ANDRADE	90
5 Evolution and Consumer Behavior VLADAS GRISKEVICIUS AND KRISTINA M. DURANTE	122
6 Consumer Neuroscience: Revealing Meaningful Relationships between Brain and Consumer Behavior HILKE PLASSMANN AND UMA R. KARMARKAR	152
7 Developmental Consumer Psychology: Children in the Twenty-First Century LAN NGUYEN CHAPLIN AND PAUL M. CONNELL	180
8 Consuming Brands JILL AVERY AND ANAT KEINAN	209

- 9 User Design through Self-Customization
CLAUDIA TOWNSEND, ULRIKE KAISER, AND MARTIN
SCHREIER 233

Part II. Interpersonal and Social Consumer Psychology

- 10 Identity-Signaling Behavior
DAVID GAL 257
- 11 Coping Research in the Broader Perspective: Emotions,
Threats, Mindsets, and More
DAHEE HAN, ADAM DUHACHEK, AND NIDHI AGRAWAL 282
- 12 Power and Consumer Behavior
DEREK D. RUCKER AND ADAM D. GALINSKY 309
- 13 Social Hierarchy, Social Status, and Status Consumption
DAVID DUBOIS AND NAILYA ORDABAYEVA 332
- 14 Word of Mouth and Interpersonal Communication
JONAH BERGER 368
- 15 Gift Giving
MORGAN K. WARD AND CINDY CHAN 398
- 16 Interpersonal Influences in Consumer Psychology:
When Does Implicit Social Influence Arise?
KIRK KRISTOFFERSON AND KATHERINE WHITE 419
- 17 Agency and Communion as a Framework to Understand
Consumer Behavior
DIDEM KURT AND JEREMY FRIMER 446
- 18 Online Social Interaction
ANDREW D. GERSHOFF AND ASHESH MUKHERJEE 476

Part III. Societal Structures

- 19 Ethical Consumption
REBECCA WALKER RECZEK AND JULIE R. IRWIN 507
- 20 Government Efforts to Aid Consumer Well-Being: Understanding
Federal Health Warnings and Disclosures
JEREMY KEES, SCOT BURTON, AND J. CRAIG ANDREWS 530
- 21 Taxes and Consumer Behavior
CHRISTOPHER Y. OLIVOLA AND ABIGAIL B. SUSSMAN 564
- 22 Moral and Political Identity
KAREN PAGE WINTERICH, VIKAS MITTAL, AND KARL
AQUINO 589

23	The Consumer Psychology of Online Privacy: Insights and Opportunities from Behavioral Decision Theory	619
	LESLIE K. JOHN	
24	Consumers and Healthcare: The Reluctant Consumer	647
	JANET A. SCHWARTZ	
25	Social Class and Scarcity: Understanding Consumers Who Have Less	673
	ANUJ K. SHAH	
26	Consumer Sharing: Collaborative Consumption, from Theoretical Roots to New Opportunities	693
	CAIT LAMBERTON	
27	Globalization, Culture, and Consumer Behavior	721
	CARLOS J. TORELLI AND SHIRLEY Y. Y. CHENG	
	<i>Name Index</i>	749
	<i>Subject Index</i>	756

Figures

6.1	Use of Neuroscientific Tools in (A) Academic Research and (B) Practitioner Research	<i>page</i> 157
6.2	Brain Areas Involved in Consumer Decision Making (Adapted from Plassmann et al. 2012)	160
12.1	A Nomological Net of the Emerging Study of Power in Consumer Behavior	310
13.1	Percentage of Articles Including One of the Search Terms in Leading Marketing, Sociology, Social Psychology, and Management Journals, February 2011–February 2014	333
14.1	Visual Depiction of Key Communication Factors	369
20.1	A Model of Consumer Responses to Warnings and Disclosures	532
20.2	Examples of Graphic Health Warnings	534
20.3	Examples of Different Types of Front-of-Package Nutrition Systems	544
20.4	Current Nutrition Facts Panel Compared to the Proposed 2014 Panel	548

Tables

5.1	Four Types of Explanations for Behavior	<i>page</i> 124
5.2	Sampling of Evolutionarily Informed Theories	128
6.1	Summary of Neuroscience Methods	154
11.1	Antecedents and Consequences of Different Coping Strategies	294
13.1	Number of Articles Including One of the Search Terms in Leading Marketing, Sociology, Social Psychology, and Management Journals, February 2011–February 2014	333
17.1	Measures of Agency and Communion	448
20.1	Overview of Some Current Warning and Disclosure Issues and Examples of Possible Research Questions	554

Contributors

NIDHI AGRAWAL, University of Washington

EDUARDO B. ANDRADE, Brazilian School of Public and Business Administration, FGV, Rio de Janeiro

J. CRAIG ANDREWS, Marquette University

KARL AQUINO, University of British Columbia

JILL AVERY, Harvard Business School

JONAH BERGER, University of Pennsylvania

PABLO BRIÑOL, Universidad Autónoma de Madrid

SCOT BURTON, University of Arkansas

CINDY CHAN, University of Toronto

LAN NGUYEN CHAPLIN, University of Illinois at Chicago

SHIRLEY Y. Y. CHENG, Hong Kong Baptist University

PAUL M. CONNELL, Stony Brook University

DAVID DUBOIS, INSEAD

ADAM DUHACHEK, Indiana University

KRISTINA M. DURANTE, Rutgers University

JEREMY FRIMER, University of Winnipeg

DAVID GAL, University of Illinois at Chicago

ADAM D. GALINSKY, Columbia University

ANDREW D. GERSHOFF, University of Texas at Austin

VLADAS GRISKEVICIUS, University of Minnesota

DAHEE HAN, McGill University

HAL E. HERSHFELD, University of California at Los Angeles

JULIE R. IRWIN, University of Texas at Austin

LESLIE K. JOHN, Harvard University
ULRIKE KAISER, Institute for Marketing Management, WU Vienna
UMA R. KARMARKAR, Harvard Business School
JEREMY KEES, Villanova University
ANAT KEINAN, Harvard Business School
KIRK KRISTOFFERSON, University of British Columbia
DIDEM KURT, Boston University
CAIT LAMBERTON, University of Pittsburgh
VIKAS MITTAL, Rice University
CASSIE MOGILNER, University of Pennsylvania
CAREY K. MOREWEDGE, Boston University
ASHESH MUKHERJEE, McGill University
MICHAEL I. NORTON, Harvard Business School
CHRISTOPHER Y. OLIVOLA, Carnegie Mellon University
NAILYA ORDABAYEVA, Boston College
HILKE PLASSMANN, INSEAD
REBECCA WALKER RECZEK, Ohio State University
DEREK D. RUCKER, Northwestern University
MARTIN SCHREIER, Institute for Marketing Management, WU Vienna
JANET A. SCHWARTZ, Tulane University
ANUJ K. SHAH, University of Chicago
ABIGAIL B. SUSSMAN, University of Chicago
CARLOS J. TORELLI, University of Minnesota
ZAKARY L. TORMALA, Stanford University
CLAUDIA TOWNSEND, University of Miami
MORGAN K. WARD, Southern Methodist University
KATHERINE WHITE, University of British Columbia
KAREN PAGE WINTERICH, Pennsylvania State University

Introduction

Understanding Consumers in the Here, the Now,
and the Tomorrow

Michael I. Norton, Derek D. Rucker, and Cait Lamberton

Study the past if you would divine the future.

– Confucius

The words of Confucius define the architecture of this handbook. As editors, we made a decision to seek out a set of scholars that we agreed had conducted some of the most exciting and novel research in their areas of expertise in recent years, and who we felt were likely to play a leading role in shaping the agenda for consumer behavior research in the years to come.

Although we left details of the chapter contents to our authors, as editors we gave all authors the same core blueprint to build the foundation of their chapters. First, we asked our authors to look backward into the literature to offer a discussion of key developments, theoretical arguments, and empirical findings to provide readers with a critical summary of a subject central to consumer psychology – with a focus on research conducted over the last decade, to make each chapter a tour of those research papers likely to become “classics” in the years to come. Second, we asked each author to set a research agenda to address unanswered questions and topics of study that they felt should become part of the future canon of their research area. As a result, while each chapter has its own particular feel and design, the volume as a whole is linked by the common motifs of both the past and future of a research area.

In this introduction, we provide a brief overview of the chapters to come. The handbook is divided and organized by three central lenses on consumer psychology: individual, social, and societal.

Chapters in [Part I](#) of the volume emphasize consumer decision making and behavior at the level of the individual. The first five chapters explore the building blocks of consumer behavior from a variety of theoretical orientations, from understanding consumer well-being (Mogilner and Norton), to persuasion (Tormala and Briñol), to predictions about the future (Morewedge and Hershfield), to emotions (Andrade), to evolutionary perspectives (Griskevicius and Durante). Two chapters focus on promising, yet underutilized, approaches to consumer psychology, from neuroscientific techniques (Plassmann and Karmarkar) to developmental approaches (Chaplin and Connell). The final two chapters discuss novel developments in two areas of marketing with rapidly changing landscapes: branding (Avery and Keinan) and co-creation (Townsend, Kaiser, and Schreier).

[Part II](#) emphasizes interpersonal and social aspects of consumer psychology. Here the focus shifts to understanding how consumer behavior is affected by the real or

even imagined presence of others. Discussions of how others are central to self-definition are introduced with implications for consumption as a means of signaling one's desired identity to others (Gal) and for coping with threat (Han, Duhachek, and Agrawal). This section also recognizes hierarchy – the ranking of individuals – as a fundamental aspect of social relations. To understand the implications of hierarchy for consumer behavior, perspectives are provided with regard to both power (Rucker and Galinsky) and status (Dubois and Ordabayeva). The section continues with an examination of social dynamics regarding the sharing of information between consumers (Berger), the giving of gifts (Ward and Chan), and interpersonal influence (Kristofferson and White). Finally, several emerging perspectives are brought to bear with regard to the role of agentic versus communal orientations (Kurt and Frimer) and online consumer interaction (Gershoff and Mukherjee).

In [Part III](#), authors focus on consumers' embeddedness in – and, often, definition in terms of – broader societal structures. Chapters suggest that rather than consisting of individually constructed identities, consumers' self-definitions are radically shaped by the cultures in which they live (Torelli and Cheng), the economic constraints they face (Shah), the political and moral forces with which they align (Winterich, Mittal, and Aquino), and the experiences of public and collective spending by which they are legally constrained (Olivola and Sussman). At the same time, consumers are not passive actors in their culture – they can choose to purchase in ways that create positive outcomes for others (Reczek and Irwin) or to pool resources in ways that lead not only to survival but to creativity and production (Lamberton). Further, they can make active decisions about their own well-being, whether physical, as in the case of healthcare decisions (Schwartz), or psychological, as they create boundaries between themselves and others (John). Finally, the ways that government works to both protect the individual and empower better decision making are explored (Kees, Burton, and Andrews).

As the chapters reveal, understanding consumers – from individual to social to societal – necessitates drawing on a wide range of theory and approach, ranging from anthropology to behavioral decision theory to neuroscience to economics to developmental psychology. Our authors have done an impressive job of beginning the process of integrating these disparate perspectives; this integration will be crucial in exploring and understanding consumer phenomena as they continue to evolve. We hope this this volume, by bridging both the past and looking to the far horizon of the future, will serve not simply as an archive but as an inspiration for research that will form the building blocks of future handbooks.

PART I

Individual Consumer Decision Making and Behavior

1 Consumer Happiness and Well-Being

Cassie Mogilner and Michael I. Norton

What has consumer research taught us about how to make our lives better? By “better,” we mean according to *us* consumers, in terms of how happy we feel in our lives and with the choices we make: our subjective sense of well-being. Having surveyed the research, it is our pleasure to report that over the past ten years, consumer research has taught us quite a lot in terms of things we can do, choices we can make, and ways that we can think to increase our happiness. Still, for as much as we have learned, there is even more yet to learn, which makes consumer happiness a ripe topic for future research.

We can attribute much of the impetus for consumer happiness research to two sources: Martin Seligman’s American Psychological Association Presidential Address in 1998 and David Mick’s Association of Consumer Research Presidential Address in 2005. Both of these speeches implored their respective fields, Seligman’s psychology and Mick’s consumer research, to shift the research agenda toward improving subjective well-being. In Seligman’s 1998 address, he redefined the mission of psychology to “making the lives of all people better,” and in doing so carved out the subfield known as Positive Psychology: “a science and profession whose aim is the building of what makes life most worth living.” In Mick’s 2005 address, he suggested that the field should pursue “investigations that are framed by a fundamental problem or opportunity, and that strive to respect, uphold, and improve life in relation to the myriad conditions, demands, potentialities, and effects of consumption,” terming this pursuit “transformative consumer research.”

This chapter reviews research over the past decade that heeded these urgings and sought to transform consumers’ lives by increasing their happiness and well-being. This research has typically taken one of two approaches: investigating the happiness consumers feel in their lives at a broad level, or investigating the happiness consumers feel with respect to more specific consumption episodes. The chapter is organized as such. To begin our review, since consumption typically relies on the expenditure of either or both of our two fundamental resources – money and time – we first discuss research (including some of our own) that relates these resources to happiness in general. For instance, we review research that addresses whether having money contributes to happiness, and research that identifies particular ways of spending money

that contribute to happiness. We also review research that examines what happens when consumers shift their attention away from money and toward time, and research that explores particularly happy ways of spending time. We wrap up this first section by highlighting work that shows the effect of feeling happy on subsequent thinking and behavior. The second section then surveys research examining the happiness that consumers feel from their choices and consumption experiences more specifically. Finally, we conclude the chapter by identifying some of the many unanswered questions that remain for future research.

Since the two presidential addresses that urged discoveries in this area, a notable body of research has emerged, filling the pages of such books as Aaker and Smith's (2010) *The Dragonfly Effect* and Dunn and Norton's (2013) *Happy Money*. And even though it keeps the two of us very busy reviewing papers and conference submissions, we are thrilled to see the increasing enthusiasm, especially among junior scholars, for this topic of consumer happiness and well-being. It is tremendously encouraging to observe the efforts within our field to provide empirical insights into the age-old question of what makes people happy.

Life Happiness

Happiness from Money

Despite people's intuitions to the contrary, it turns out that more money does not tend to be associated with much more happiness, particularly after people's basic needs are met (Aknin, Norton, & Dunn, 2009; Kahneman & Deaton, 2010; Kahneman, Krueger, Schkade, Schwarz, & Stone, 2006). To be sure, more money has not been shown to make people *less* happy – it is simply the case that the relationship tends to be modest (Diener & Biswas-Diener, 2002; Frey & Stutzer, 2000). But do these correlational studies suggest that money cannot buy happiness, or rather that people are not using their money in ways that maximize their “happiness bang for the buck”? Recent research suggests that when people shift from spending their money on material possessions to other categories – such as prosocial spending and buying experiences – money can in fact pay off in more happiness.

Prosocial Spending

What would make consumers happier – receiving a \$5 windfall and being instructed to spend it on themselves, or receiving that same \$5 windfall and being asked to spend it on someone else, say, by taking a friend to lunch, or donating to charity? Despite people's strong belief that spending on themselves would make them happier, research shows that people who engage in prosocial spending are reliably happier than those who spend on themselves

(Dunn, Aknin, & Norton, 2008; Hill & Howell, 2014; for a review, see Dunn, Aknin, & Norton, 2014). Moreover, people who chronically spend on others are happier people overall: the more people give, even controlling for how much money they make, the higher their well-being (Dunn, Aknin, & Norton, 2008). These results are not limited to affluent consumers in the United States. Indeed, in countries ranging from Canada to Uganda, from India to South Africa, spending on others is both correlated with happiness and causes people to become happier (Aknin et al., 2013). Happiness from giving even occurs among toddlers – a consumer segment seemingly full of “mine-o-saurs” whom we do not expect to be particularly generous (Aknin, Hamlin, & Dunn, 2012).

Why does prosocial spending pay off more than personal spending? Several studies offer hints by identifying critical moderators that enhance the emotional benefits of giving. First, spending on strong ties leads to more happiness than spending on weak ties. Although both lead to more happiness than spending on the self, spending on a friend tends to lead to more happiness than spending on a charity (Aknin, Sandstrom, Dunn, & Norton, 2011). Relatedly, seeing the recipient of one’s generosity increases the happiness gained from giving, which underscores the “social” value of prosocial spending (Aknin, et al., 2013). So, are charities and non-profits doomed to have unhappy donors? Luckily, research suggests another critical moderator: the feeling of having had a specific impact. As a result, framing charitable donations as impacting a specific recipient leads to more happiness (Aknin, et al., 2013).

Buying Experiences

A growing set of findings in psychology and consumer research advises that despite the allure of acquiring material possessions, people should acquire experiences to enjoy greater happiness (for a review, see Gilovich, Kumar, & Jampol, in press). Researchers have shown that positive experiences tend to bring greater enduring happiness than positive material purchases (Howell & Hill, 2009; Nicolao, Irwin, & Goodman, 2009; Van Boven & Gilovich, 2003). Compared to material purchases, experiential purchases are more self-defining (Carter & Gilovich, 2012), more interpersonally connecting (Chan & Mogilner, 2014), harder to compare against forgone alternatives (Carter & Gilovich, 2010), more unique (Rosenzweig & Gilovich, 2012), and subject to slower rates of hedonic adaptation (Nicolao, Irwin, & Goodman, 2009). Indeed, even the experience of waiting for an experience is superior to that of waiting for a material good: whereas the former feels like excitement, the latter feels more like impatience (Kumar, Killingsworth, & Gilovich, 2014). Lastly, shared experiences tend to be more enjoyable than solitary experiences (Caprariello & Reis, 2013), suggesting – as with spending on others – the hedonic benefits of using money to create social connection.

Happiness from Time

Thinking about Time

In his famous adage, “Time is money,” Benjamin Franklin equates time to money as a scarce resource. Like money, time can be saved, budgeted, wasted, and spent. Although a considerable amount of research in the fields of marketing and psychology focuses on money, dealing with questions of price, willingness to pay, materialism, and wealth; and although searches on Google reveal consumers to be similarly absorbed with money (Gino & Mogilner, 2014), recent research argues for a shift in attention toward time by showing the positive downstream consequences of thinking about time rather than money. This work doesn’t look at the effect of *having* money or *having* time, but instead at the arguably broader and more controllable impact of merely thinking about one resource versus the other.

In her 2010 paper, Mogilner demonstrated that directing attention to time instead of money increases individuals’ happiness (by motivating them to socialize more and work less). This effect was shown among typical Americans and low-income Americans planning their days, and among café patrons’ visiting a local coffee shop. In this latter experiment, people entering a café were asked to take part in a questionnaire that involved unscrambling a series of sentences that surreptitiously exposed them to time-related words, money-related words, or only neutral words. Unbeknownst to them, these participants were then observed to see how they spent their time in the café (i.e., socializing with fellow patrons or doing work), and when exiting the café were asked to complete a second questionnaire about how happy and satisfied they felt. Those who had been primed with time upon entering the café spent a greater proportion of their time socializing and left happier, whereas people who had been primed with money spent more time with their noses buried in books and left less happy. So, even though focusing on money motivates people to work more (which is useful to know when looking for that extra push to meet a looming deadline), passing the hours working does not translate into greater happiness. Spending time with loved ones does, and a shift in attention toward time proves an effective means to motivate this happy behavior.

Gino and Mogilner (2014) later tested how focusing on time (vs. money) influences morality, and in so doing provided insight into the mechanism underlying the positive effect of thinking about time on happiness. Whereas thinking about money makes people more likely to cheat, thinking about time makes people less likely to cheat, which happens because thinking about time pushes people to reflect more on themselves. The researchers showed this through both the mediating role of self-reflection and the presence of a mirror, which elicited the same effect as priming time. Thus, drawing attention to time seems to nudge people to consider their time in life as ultimately finite, which encourages them to act in ways they can be happy with when holding up this figurative mirror to who they are.

The self-reflecting quality of time also impacts the happiness consumers enjoy from their products. Activating the concept of time leads consumers to be both more likely to purchase and more likely to enjoy products than when the concept of money is activated. Mogilner and Aaker (2009) demonstrated this effect among customers purchasing a cup of lemonade from a lemonade stand, university students thinking about their iPods, restaurant patrons asked about eating out, and typical consumers evaluating their cars. As an example, students first asked how much time they had spent on their iPod liked their iPod significantly more than students who were first asked how much money they had spent on their iPod. This was because thinking about their temporal investment made the students feel that the product said more about who they are than did thinking about their financial investment. (Notably, activating money leads to a more favorable effect than activating time for prestige possessions and highly materialistic consumers – where instead of one's experience with the product, one's possession of the product is a clearer reflection of the self.) The benefits of directing consumers' attention to time rather than money have also been documented in the context of charitable giving (Liu & Aaker, 2008). Because thinking about giving time to a charitable cause evokes greater happiness than thinking about giving money to that cause, donation solicitations that first ask for time produce greater donations – both in the form of time and money.

Not only can thinking about the broad construct of time influence happiness, but so can the *way* people think about time. In particular, how much time people feel like they have left in life and their associated appreciation of the present moment influence how happiness is experienced. Mogilner, Kamvar, & Aaker (2011) studied the emotions expressed on millions of personal blogs in order to understand how people experience feeling happy. This analysis, along with a series of laboratory experiments and surveys, identified two primary forms of happiness: one associated with feeling excited and the other associated with feeling calm. Moreover, through using the bloggers' profile information, the researchers could examine who was expressing each type of happiness and identified a gradual, age-based shift, such that people in their teens and twenties were more likely to experience happiness as excitement, but as people got older, they became more likely to experience happiness as calm.

These researchers' subsequently dug into the psychology underlying this shift and examined how it plays out to influence the choices consumers make (Mogilner, Aaker, & Kamvar, 2012). They concluded that because younger people typically have a longer future ahead of them, the more future-oriented emotion of excitement tends to feel more positive in its alignment with these youngsters' eagerness for that future. However, as people get older, their future becomes more limited and calm starts to feel more positive in its alignment with these more "mature" adults' appreciation of the present moment. Therefore, in pursuit of products that will make them feel happy, younger consumers (or those who eagerly await their futures) are more likely to choose exciting options, whereas older consumers (or those who appreciate the present) are

more likely to choose calming options. Notably, an individual's temporal orientation toward the future or present can be manipulated such that young people can instead pursue and experience happiness from calm, and older people can instead pursue and experience happiness from excitement. This means that although the way people experience happiness is dynamic and naturally shifts over the course of life, it is also malleable. Furthermore, whether people experience happiness more from excitement or calm plays into the choices they make, such as the music they listen to, the type of tea they drink, and the brand of water they buy.

Even though young people generally view the remainder of their lives to be relatively long (Carstensen, Isaacowitz, & Charles, 1999), when asked the question, "Is life short or long?" the majority of people (irrespective of age) answer that life is short. They also tend to believe that life is hard rather than easy, which puts the majority of people in agreement with Hobbes's view of life as "nasty, brutish, and short." Norton, Anik, Aknin, and Dunn (2011) tested whether such a philosophy of life relates to happiness. They asked people two binary choice questions – "Is life short or long?" and "Is life easy or hard?" – and measured their happiness. The results revealed that holding the short-hard philosophy is associated with less happiness compared to a long-easy philosophy. Larsen and McKibban (2008) investigated another philosophy proposed by Rabbi Hyman Schachtel in 1954: "Happiness is not having what you want, but wanting what you have." To test this maxim, the researchers asked people whether or not they had, and the extent to which they wanted, fifty-two material items. This allowed the researchers to quantify how much their participants wanted what they had and how much they had what they wanted. The results showed that both variables had distinct and positive effects on happiness, and wanting what you have helps explain the positive effect of gratitude on happiness.

Together these findings suggest that focusing on the broad notion of time increases happiness and that thinking about time in a particular way (as long or short, and with respect to the future or the present) can influence not only how much happiness people feel but also how they experience happiness.

Spending Time

How does spending time influence happiness, and are there particular ways of spending time that produce greater happiness? A group of papers examining the former showed that investing time in product creation can increase how happy consumers subsequently are with that product. Two papers by Norton, Mochon, and Ariely in 2012 uncovered what they called the "IKEA effect." Their findings showed that individuals who built IKEA boxes, folded origami frogs, or constructed sets of Legos were willing to pay more for their finished product than were individuals who had been presented with these same products already assembled by an expert. Whether a self-proclaimed "do-it-yourselfer" or not, it turns out that spending time to successfully assemble a product makes

people feel competent and accomplished. The sense of accomplishment tied to these self-assembled products drives people's value for the products. A similar "I designed it myself" effect uncovered by Franke, Schreier, and Kaiser (2010) shows that the benefits of product assembly also pertain to product design. Individuals who designed their own T-shirt or set of skis were willing to pay more for these products than individuals presented with "off-the-shelf" versions of these products. This may not initially seem all that surprising, since designing one's own T-shirt allows consumers to customize their shirts to fit their individual tastes; however, the researchers showed that participants were willing to pay more for a T-shirt that they had designed themselves, even when it looked (and was) exactly the same as the off-the-shelf version. As with the IKEA effect, an increased sense of accomplishment played an underlying role. In both cases, investing time into a product provides feelings of accomplishment, which afford consumers greater happiness from these products. Buechel and Janiszewski (2014) more recently noticed that such benefits may be specific to cases in which design and assembly are combined. When consumers are tasked with designing and assembling their products but these activities are separated, the time that consumers are required to spend becomes onerous. In related work, Moreau, Bonney, and Herd (2011) showed that customization of products is particularly positive when the product is intended as a gift for someone else. In a study among women tasked with designing a tote bag either for herself or for a friend using a real customization website, the researchers found that the women who created their tote bag as a gift were willing to pay significantly more than those who created their tote bag for themselves.

Both the IKEA effect and the "I designed it myself" effect were driven by an increased sense of accomplishment from spending time on a product. This happiness from "doing" may be tapping into something even more fundamental. Hsee, Yang, and Wang (2010) found that although people tend to choose to remain idle, busy people are happier than their idle counterparts. The researchers found across two studies that among college students who had a spare fifteen minutes between two surveys, those who spent the time walking to deliver their first survey to a faraway location felt happier than those who delivered their first survey to a nearby location and idly waited for the next survey to begin.

Mogilner, Chance, and Norton (2012) also noted that despite people's inclination not to exert themselves in their temporal expenditures, doing so can have surprisingly positive ramifications. In particular, although time-constrained people tend to be stingy with their time, if they were to instead give some of their time away, they might feel *less* constrained by time. Four experiments showed that individuals who spent between five and thirty minutes on someone else subsequently felt as though they had more time than did individuals who had wasted that time, spent that time on themselves, or received that time as a windfall. One study, for instance, asked participants to recall a recent expenditure of time that was either for someone else or for themselves. The researchers then asked these individuals how much time they felt like they had and how that expenditure made them feel. The results

indicated that although giving time made participants feel more connected to others, it was actually their increased sense of efficacy that was responsible for the positive effect on time affluence. Spending time to help others makes people feel as though they have accomplished more within a given amount of time, such that they can do more with their time in general. This makes time seem more expansive. Therefore, when feeling time-constrained, people should become more generous with their time, despite their inclinations to be less so.

While these papers speak to the potential happiness that can come from investing one's time into products and other people, there is also work that has looked at whether there are temporal expenditures that tend to be happier in and of themselves. Do some experiences tend to be happier than others? Keinan and Kivetz (2011) found that even though staying at a freezing ice hotel or eating such a peculiar food as bacon ice cream may not be particularly pleasant in the moment, people are motivated to collect unusual and extreme experiences in an effort to build their "experiential CV." Additionally, Zauberman, Ratner, and Kim (2009) found that people are motivated to protect their special experiences by not repeating them. Bhattacharjee and Mogilner (2014) examined the happiness derived from different types of experiences more directly. They compared the happiness associated with extraordinary experiences (those that are infrequent and go beyond the realm of everyday life – like incredible vacations and life milestones, such as graduations and weddings) and ordinary experiences (those that are frequent and within the realm of everyday life – like indulging in a delicious treat or having lunch with a friend). Bhattacharjee and Mogilner measured the happiness produced by over 1,800 self-reported experiences that people recalled, imagined, or posted on Facebook and found that the amount of happiness enjoyed from ordinary and extraordinary experiences depended on people's age – more specifically, how much time they felt they had left in life. Whereas younger people enjoyed greater happiness from extraordinary experiences, ordinary experiences became increasingly associated with happiness as people got older, such that ordinary experiences produced as much happiness as extraordinary experiences when people felt they had limited time remaining. An exploration into the multiple factors that might explain this age-based effect produced evidence for the underlying role of self-definition. Irrespective of age, self-defining experiences produce greater happiness, but the type of experiences people use to define themselves seems to change with age. Whereas extraordinary experiences are particularly self-defining for young people, older people are more likely to define themselves using the ordinary things that fill their daily lives.

By measuring people's happiness immediately before and after their attendance of religious services and going to the gym, Mochon, Norton, and Ariely (2008) also noted the potential happiness from frequent, ordinary experiences. They found not only that people were happier exiting the experience than entering it, but the frequency with which people engaged in these experiences was positively related to their overall happiness. Through frequent positive

boosts, these seemingly minor experiences promise a long-lasting impact on happiness.

Aaker, Rudd, and Mogilner (2011) proposed five principles of happy ways to spend time to complement Dunn and Norton's (2013, and Dunn, Gilbert, & Wilson's, 2011) principles of happy ways to spend money. These principles include spending time on the right activities, expanding your time, and being aware that happiness changes over time.

Effects of Happiness

Having discussed factors that influence happiness, we now turn to factors that are influenced by happiness. A review paper in 2005 in *Psychological Bulletin* by Lyubomirsky, King, and Diener delineated results revealing happiness to have positive consequences across our three domains of life: work, interpersonal relationships, and health. For instance, in the workplace, happiness was found to make employees more creative, more collaborative, and more flexible in their problem solving. In social settings, happiness was found to make people more liked by others and more liking of others, more outgoing and in-tune with others' feelings, and more prosocial with an increased tendency to volunteer. In the health domain, happiness was found to increase people's thresholds for pain, to make them more likely to follow treatment regimes, and to increase immune functioning. This list includes just experimental results that allow for a causal inference, but the review also included myriad correlational and longitudinal results, which further identify the benefits of feeling happy, such as living longer.

There have since been more papers investigating how happiness influences consumers' thinking and behavior. A nice example by Labroo and Patrick (2009) shows how being in a good mood elevates people's thinking so that they see the big picture. They reasoned that being in a positive mood signals the situation to be benign, which allows people to distance themselves psychologically from the situation. Feeling happy thus increases abstract construal and high-level thinking. On the other hand, a negative mood signals that danger may be imminent. This leads people to take a more proximal perspective that results in a concrete low-level construal. Indeed, participants who were made to feel happy (either through something as subtle as exposure to smiley-face bullet points or as vivid as intricately describing the happiest day of their life) were more likely to (1) generate abstract descriptions of activities (e.g., painting a room was described as beautifying the environment rather than choosing a favorite paint color), (2) endorse abstract goals (e.g., studying for an exam was more important when thinking about why they should study rather than how they should study), and (3) prefer products advertised with abstract messages (e.g., an orange juice was better liked when framed as helping you "invest in your future health" rather than "ensure your health today").

Happiness influences not only the way consumers think but also the way consumers behave. For instance, people put in a good mood were more likely to prefer the first option presented to them (except when instructed to withhold their judgments until all options had been presented, in which case they preferred the last option (Qiu & Yeung, 2008), to perform worse on tasks subject to predecisional distortion or overconfidence (Meloy, Russo, & Miller, 2006), to engage in emotion regulation when holding the belief that emotion is fleeting (Labroo & Mukhopadhyay, 2009), and to be less suspicious of persuasion attempts (DeCarlo & Barone, 2009). Notably, not all of these are positive outcomes.

In an effort to identify ways to increase consumers' happiness, the bulk of the research covered in this chapter treats happiness as the dependent variable. The research covered in this section, however, treats happiness as the independent variable to identify the consequences of feeling happy. A recent paper by Aknin, Dunn, and Norton (2012) took yet another approach and employed happiness as both the independent variable and the dependent variable, which allowed them to identify a positive feedback loop between prosocial spending and happiness. With their prior work showing that spending money on others boosts happiness, here they manipulated happiness by instructing half of their participants to recall an occasion in which they had spent money on others. All participants then reported their happiness and their propensity to spend money on others in the future. The results offered a potential path to sustainable happiness whereby prosocial spending increases happiness, which in turn encourages the happy behavior of spending prosocially. This suggests that one potential outcome of happiness is more happiness.

Consumption Happiness

Although consumers all strive to live happy lives, this is admittedly a fairly lofty pursuit. Consumer researchers have therefore tended to instead focus on understanding the more specific and perhaps attainable goal of feeling happy with the individual choices consumers make and the more discrete experiences that comprise consumers' lives. In this portion of the chapter, we highlight recent work that examines factors that influence the happiness consumers enjoy from their choices and experiences.

Choice Satisfaction

A longstanding and important literature has shown that people *want* choice (e.g., Deci & Ryan, 1985; Ryan & Deci, 2000). Due to the sense of freedom and autonomy that choosing affords, outcomes that are personally selected are often associated with greater task enjoyment and evaluations (Langer 1975; Langer & Rodin 1976). More recently, however, there has been a growing stream of research exploring *when* choosing leads to greater happiness with

one's chosen outcome, as well as when the responsibility of choosing can be taxing and lead to *less* happiness. For instance, Botti and McGill (2006) found that when consumers cannot perceive differences between their options, choosing has little effect on their ultimate choice satisfaction. Indeed, consumers have a particularly difficult time perceiving variety within their choice set when there is an overwhelmingly large number of options (Mogilner, Rudnick, & Iyengar, 2008). However, when consumers are able to see the differentiation among their options, choosing increases satisfaction with one's choice when selecting from a set of positive outcomes (e.g., gourmet coffees or chocolates), but decreases satisfaction with one's choice when selecting from a set of negative outcomes (e.g., gross smells; Botti & McGill, 2006). Unless consumers see their options as differentiated, they do not feel as though they have really made a choice and thus do not enjoy the credit (or suffer the blame) from arriving at a positive (or negative) outcome.

Although consumers typically insist on exercising choice, choosing can lead to less happy outcomes. Botti and Hsee (2010) showed this among undergrads presented with ten multiple choice questions from the quantitative section of the General Records Examinations (GRE). Participants were given twenty minutes and told that they would get 40 cents for each correct answer, lose 10 cents for each incorrect answer, and get 20 cents for each minute remaining after they had answered all ten questions. Half of the participants were allowed to choose which ten questions out of a larger set of fifty they would answer, whereas the other half were assigned their ten questions. Students who were free to choose their questions spent more time, felt worse while taking the test, and were not more accurate than those who had not been given a choice. Still, additional studies showed that participants expect to enjoy tasks more and to do better on those tasks when given the freedom to choose, which suggests that consumers' preference for choice may partly stem from their tendency to underweight the temporal costs of searching for the best option relative to the benefits provided by that option.

A set of studies conducted by Mogilner, Shiv, and Iyengar (2013) similarly showed that consumers' satisfaction with whatever they choose can be undermined when consumers hold out hope for a more perfect option. Indeed, comparing the satisfaction and commitment levels between choosers presented with their options simultaneously (all at once) or sequentially (one at a time), the researchers found that sequentially presented options elicit a focus on the future and hope for a better option to come, which leaves choosers less satisfied with and ultimately less committed to whichever option they had selected. Although this effect was demonstrated among individuals choosing a piece of gourmet chocolate, a nail polish color for a salon manicure, or a bottle of Italian red wine from a wine tasting, it has implications for individuals who will likely be presented with their options one at a time for such critical outcomes as a job, house, or spouse. These findings advise that to enjoy greatest happiness from choice, consumers should give up hoping for what is to come and remain

focused on that which they currently have. Gu, Botti, and Faro (2013) offer an easy method by which to psychologically conclude the decision-making process so as to increase satisfaction with one's choice: do a physical act that communicates "choice closure," like closing the menu after deciding which option you'll order.

Although consumers think they want choice – preferring to compare their options side-by-side (Hsee & Zhang, 2004) and to be fully informed about the specifications of their options (Hsee, Yang, Gu, & Chen, 2009) – consumers may end up happier with their chosen outcome if they had evaluated each option separately and were not provided with specification information. Yang, Hsee, Liu, & Zhang (2011) showed this among college students who chose between two bath towels: one that looked nicer and one that was 7 millimeters thicker. Since participants would be using their chosen towel for a week in their communal dorm, and millimeter differences in towel thickness are difficult to subjectively assess, a towel's appearance should be more critical to students' happiness while using the towel than its thickness. Indeed, when asked a week later, the students who had selected the better-looking towel were happier with their chosen towel than those who had selected the thicker towel, but the better-looking towel was less likely to have been selected when the thickness specifications were presented at the time of choice and when the two towels had been presented side by side.

Research has also shown that the size of an assortment and the way it is categorized influences consumers' happiness with their chosen option, irrespective of the objective outcome. For instance, Diehl and Poynor (2010) built on prior literature that showed overly large assortments to reduce consumers' happiness with their choices (Chernev, 2003; Gourville & Soman, 2005; Iyengar & Lepper, 2000; Lehmann 1998) by identifying an explanation for why this choice overload effect might occur. It has to do with expectations. When selecting a birthday card for a colleague, those who chose from one hundred cards had higher expectations to be able to precisely match their card preferences than did those who chose from only ten cards. The disconfirmation of their high expectations resulted in choosers in the one hundred assortment size condition to be less satisfied with their chosen birthday card than choosers in the ten assortment size condition (Diehl & Poynor, 2010). Fortunately for consumers, Mogilner, Rudnick, and Iyengar (2008) showed that such choice overload can be offset through option categorization. The researchers found that consumers selecting a magazine from a vast supermarket magazine aisle or a cup of coffee from an extensive menu of blends were more satisfied with their chosen option when the assortment was divided into more (vs. fewer) categories, regardless of whether the category labels contained information. This "mere categorization effect" was stronger among consumers unfamiliar with the choice set – those who were unable to perceive differences among the vast array of options without some help. Poynor and Wood (2010) later found that the *way* options are categorized can also have an effect. For knowledgeable consumers, unexpected categorization provides a newness cue, which engages them in the decision process,

thereby increasing their satisfaction with their chosen option. For less knowledgeable consumers, however, such unexpected categorization reduces choosers' satisfaction.

All of these papers highlight how one's experience during the decision-making process can influence how happy one is with his or her choice, regardless of the objective qualities of that outcome. Even after the choice has been made and the decision-making process is complete, the way consumers engage with products can also influence their happiness with those products. Vohs, Wang, Gino, and Norton (2013) found that when consumers perform a ritual before consuming a product, they enjoy the product more. They showed this effect among students consuming such tasty products as chocolate and lemonade, as well as the more healthful snack of carrots. It seems that enacting a ritual draws people further into the consumption experience, such that they are more involved and savor the product more. As noted previously, investing time into products and attending to that time spent makes consumers feel more connected to those products, and thus happier with them (Mogilner & Aaker, 2009).

Experience Enjoyment

A growing body of work identifies factors – such as uncertainty, segregation of positive and negative features, and the presence of other consumers – that influence how much happiness consumers enjoy from a consumption experience.

Although people are driven to understand events in order to make them more predictable and replicable, increased understanding can also make these events less enjoyable. That is to say that *uncertainty* can increase enjoyment. Norton, Frost, and Ariely (2007) demonstrated this in what they called the “less is more effect.” In one of their studies, online daters were given a list of traits describing a potential person to date and were asked how much they liked the person. The number of traits presented varied between one and ten. Although daters believed that more information would lead to greater liking, the study results showed the opposite relationship. Knowing more led to liking less. Uncertainty also benefits romance with respect to knowing how a potential partner feels in return. Whitchurch, Wilson, and Gilbert (2010) found that people like their crush more when they are uncertain about whether their crush likes them back. Uncertainty apparently holds people's attention and keeps them from fully making sense of and adapting to a particular outcome. Even outside the realm of dating, this logic helps explain why uncertainty prolongs the enjoyment of experiences. For instance, students who received a gift of \$1 were happier when the description on the gift card was hard to figure out, and movie viewers enjoyed a film for longer when they were unsure of which of two alternate endings was true (Wilson, Centerbar, Kermer, & Gilbert, 2005). Uncertainty is particularly positive when a potential positive outcome is easy to imagine, such as winning a lucky draw (Lee & Qiu, 2009). Even though the human mind is designed to reduce uncertainty to help make sense of the world, maintaining

some degree of uncertainty allows consumers to enjoy their positive experiences for a bit longer.

Experience enjoyment is not only influenced by projecting forward but also by reflecting back. When consumers have the chance to reflect on their past experiences, they are motivated to frame them in a positive light. Cowley (2008) documented this tendency, which she termed “hedonic editing.” She found that gamblers use this to justify their behavior, and it entails segregating large gains from negative experiences (i.e., focusing on the silver lining) and combining large losses with large gains from positive experiences (i.e., hiding the less than fully pleasant truth). More generally, the way consumers remember their experiences determines their continued enjoyment.

Lastly, the extent to which consumers enjoy an experience is largely influenced by others – others who selected the experience and others who shared in the experience. With respect to the former, Chan and Mukhopadhyay (2010) showed that when choosing an experience for oneself (e.g., which of two concerts to attend), evaluation of the chosen experience initially increases, but then levels out over time. On the other hand, when someone else chooses that same experience for you, any delay leads to less favorable evaluations. With respect to the latter, Caprariello and Reis (2013) suggest that shared experiences lead to greater happiness than solo experiences. They specifically found that experiences are more likely to be shared than material possessions, which they argued explains the greater happiness produced by experiential than material purchases. Whether sharing an experience leads to greater happiness, however, depends on if those sharing in the experience see themselves as in agreement with each other (Raghunathan & Corfman, 2006). Whereas congruence in opinions enhances the enjoyment of the shared experience (both by engendering a feeling of belonging and increasing confidence in the accuracy of one’s views), incongruence in opinions diminishes such enjoyment. Ramanathan and McGill (2007) similarly found that consumers’ enjoyment of an experience moves in accordance with others’ enjoyment when they are able to observe the others’ facial expressions. Furthermore, the resulting coherence leads to more positive retrospective evaluations of the experience. Thus, it is not just whether an experience is shared that determines enjoyment, but also the extent to which the level of enjoyment is shared.

Where to Go from Here

While the body of research we have reviewed here offers a glimpse into the immense progress that has been made in understanding the psychology of consumer happiness and well-being, the relative newness of the field means that many critical, interesting, and generative questions remain to be explored. In the following subsections, we offer several avenues that we believe are likely to be fruitful – though, of course, our list is far from comprehensive.

Varieties of Well-being

The focus of much of the research we have reviewed is on happiness – often assessed using a simple scale that asks consumers to state their happiness from “not at all” to “a great deal” (or something similar). Of course, a construct of such centrality to human life as *happiness* is bound to be more multifaceted and complex. Are people looking to maximize the happiness they feel in the moment, or are they looking to maximize the happiness they feel as they reflect back on an experience or their life overall (Mogilner & Norton, 2015)? Although the behaviors that contribute to each will frequently be the same, there are important cases where they diverge, thus pitting the “experiencing self” against the “remembering self” (Kahneman, 2011). Going forward, researchers should more specifically measure each in order to identify consumption behaviors that differently influence these temporally distinct forms of happiness. Another path is to start deconstructing our understanding of what “happiness” means to consumers. One step in this direction can be seen in the research by Mogilner, Kamvar, and Aaker (2011), which demonstrates how different consumers experience happiness differently, preferring either calm or excitement. Relatedly, Rudd, Vohs, and Aaker (2012) assessed and manipulated a specific positive emotion – awe – offering a compelling example of how drilling more deeply into one aspect of happiness offers insight into consumer behavior.

It is also important to remember that feeling happy is but one aspect of overall well-being. Baumeister, Vohs, Aaker, and Garbinsky (2013) highlight meaningfulness as another. While happiness is a critical element of consumer well-being, consumers’ search for meaning and purpose deserves far more attention. Diener and colleagues have further proposed *flourishing* and *thriving* as two forms of a more encompassing construct of well-being (Diener et al., 2010; Su, Tay, & Diener, 2014). It is likely that each will be affected by different consumption experiences. Therefore, another area for future research is to begin to unpack the different aspects and varieties of consumer well-being in the hopes of developing a more complete view.

Aside from identifying specific constructs to explore, recent research also suggests that a different approach to emotion – one that focuses not on one emotion but on the range and variety of emotions that consumers experience – may also offer novel insight into consumer well-being. Certainly, consumer behavior researchers have examined consumers’ preferences for product variety (e.g., Simonson, 1990) and the ways in which product variety affects consumption and satiation (e.g., Redden, 2008), but the effect of diversity and variety of emotions on consumption has yet to be explored. Simply put, what movie or product is more desirable: one that elicits many emotions or one that elicits few? Research suggests that for overall well-being, greater emodiversity – the variety and relative abundance of the emotions people experience – is an independent and integral component of the human emotional ecosystem that predicts both mental and physical health (Quoidbach et al., 2014). Interestingly, this research

suggests that greater diversity of negative emotions also improves well-being, such that it is better to feel sad and angry than simply angry. Future research should explore the implications of emodiversity for consumer satisfaction and happiness.

Building on this question of the role of variety, future research should look to uncover optimal ways for consumers to piece together their behaviors and experiences. We reviewed recent work that identifies which individual experiences are more conducive to happiness (Bhattacharjee & Mogilner, 2014; Caprariello & Reis, 2013). Although there is still much more that can be done on that question alone, another approach is to identify which combinations of experiences are particularly conducive to happiness. For instance, Etkin and Mogilner (2015) have started to investigate whether filling one's day with a greater variety of activities leads to more or less happiness. They have found that whereas variety over the course of a day (or week, or month) increases happiness, incorporating more variety into shorter time intervals has the reverse effect, and instead reduces happiness because of a thwarted sense of progress. Relatedly, Shah and Alter (2014) have found that the way consumers categorize a series of experiences influences their sense of progress, which compels different strategies for categorizing positive and negative events. Such insights into the best ways for consumers to schedule their time offer a promising avenue through which to educate consumers on things that they can actively implement in their lives to increase their daily well-being. A good example is the work by Sellier and Avnet (2014), which shows that people who follow an event-time scheduling style (where tasks are organized based on their order of completion) are better able to savor positive experiences than people who follow a clock-time scheduling style (where tasks are organized based on a clock).

Well-being from Consumer Behaviors

In addition to delving deeper into the impact of different aspects of well-being and the role of variety, consumer behavior researchers should more carefully consider how many of the typical consumer behaviors under investigation – such as word of mouth, social networking, and conspicuous consumption – affect not only behavior but also consumer emotions. In other words, in addition to documenting ways in which various factors increase consumers' propensity to consume, researchers should consider how those consumption decisions affect consumer happiness – for better and for worse – in an effort to design marketing interventions that do not merely change behavior but also increase happiness.

Social Networking and Word of Mouth

Researchers studying word of mouth – both online and offline – have primarily been interested in understanding the factors that increase the

likelihood that a piece of information will be passed along to others (e.g., Godes & Mayzlin, 2004, 2009). At the same time, however, some research suggests that usage of online social networking sites is associated with declines in consumer well-being (e.g., Kross et al., 2013). As a result, consumer behavior researchers should be more careful in balancing whether interventions that increase word of mouth – and time on social networking sites – may be decreasing consumer happiness. Recent research hints at the power of emotion in online social networks; for example, the emotional valence of online content shapes its likelihood of being passed on to others (Berger & Milkman, 2012), and consumers' Facebook newsfeeds can be manipulated to impact consumer happiness (Kramer, Guillory, & Hancock, 2014). We suggest that researchers should search for interventions that both increase desired marketing outcomes (such as increased engagement) as well as consumer well-being. One such opportunity arises from recent research documenting the emotional benefits of matchmaking – of connecting two unacquainted people to each other (Anik & Norton, 2014). By encouraging users of online social networks to play matchmaker, marketers may be able to increase engagement and network density while simultaneously improving well-being.

Conspicuous Consumption

Consumer researchers have become increasingly interested in issues of power and status as shapers of consumer behavior (for a review, see Chapter 12 in this volume). In one intriguing set of studies, for example, Ordabayeva and Chandon (2011), showed that – in contrast to existing theory – consumers are most likely to engage in conspicuous consumption when in environments where most of their neighbors have similar possessions. In such cases, one purchase has the power to leapfrog a consumer ahead of more neighbors, making conspicuous consumption a more effective means of attaining status. In another investigation, consumers placed near the bottom of an income distribution displayed “last place aversion,” becoming more likely to gamble in an effort to escape being in last place (Kuziemko, Buell, Reich, & Norton, 2014; see also Haisley, Mostafa, & Loewenstein, 2008). And in Bellezza, Gino, and Keinan (2014), consumers selectively chose to dress down in some contexts in order to demonstrate their higher status. But what are the emotional consequences of such status-seeking behaviors? A large body of research suggests that social comparison, status seeking, and inequality can be detrimental to emotional health (e.g., Adler, Marmot, McEwen, & Stewart, 1999; Wilkinson & Pickett, 2010). Researchers should therefore examine how raising status concerns for consumers affects not only their consumption decisions but also their well-being. One early and notable example is the work by Rucker, Dubois, and Galinsky (2011), which examines the impact of different levels of power not only on consumers' spending decisions but also on their subsequent happiness.

Conclusion

The seventeenth-century philosopher and mathematician Blaise Pascal observed that “All men seek happiness. Whatever different means they employ, they all tend to this end.” This motivation to be happy is readily observable among consumers centuries later. Quite simply, consumers *want* to be happy. We are proud to be members of a field that is beginning to dedicate its efforts and talents to empirically inform this fundamental human pursuit. We hope that Seligman and Mick are similarly proud to see that their calls for such research have been enthusiastically answered. Over the past decade, a notable body of work has emerged that not only documents the different means consumers tend to employ in their search for happiness but also helps guide consumers to their desired end. Note, however, that even though tremendous strides have been made, there is much more to be done. Because consumer experience and the experience of happiness, let alone well-being more broadly, are so complex (and some would say nebulous), every compelling finding produces even more questions. We have mentioned a mere sampling of such questions, and only hope that this chapter serves to spark further interest, efforts, and questions pertaining to the meaningful improvement of consumer happiness and well-being.

References

- Aaker, J., Rudd, M., & Mogilner, C. (2011). If money does not make you happy, consider time. *Journal of Consumer Psychology*, 21(2), 126–130.
- Aaker, J. & Smith, A. (2010). *The Dragonfly Effect*. San Francisco, CA: Jossey-Bass.
- Adler N. E., Marmot, M., McEwen, B. S., & Stewart, J. (1999). *Socioeconomic Status and Health in Industrial Nations: Social, Psychological and Biological Pathways*. New York: Ann NY Acad Sci.
- Aknin, L. B., Barrington-Leigh, C. B., Dunn, E. W., Helliwell, J. F., Burns, J., Biswas-Diener, R., Kemeza, I., Nyende, P., Ashton-James, C. E., & Norton, M. I. (2013). Prosocial spending and well-being: Cross-cultural evidence for a psychological universal. *Journal of Personality and Social Psychology*, 104(4), 635–652.
- Aknin, L. B., Dunn, E. W., & Norton, M. I. (2012). Happiness runs in a circular motion: Evidence for a positive feedback loop between prosocial spending and happiness. *Journal of Happiness Studies*, 13(2), 347–355.
- Aknin, L. B., Dunn, E. W., Sandstrom, G. M., & Norton, M. I. (2013). Does social connection turn good deeds into good feelings? On the value of putting the “social” in prosocial spending. *International Journal of Happiness and Development*, 1(2), 155–171.
- Aknin, L. B., Hamlin, J. K., & Dunn, E. W. (2012). Giving leads to happiness in young children. *PLoS One*, 7(6), e39211.
- Aknin, L. B., Norton, M. I., & Dunn, E. W. (2009). From wealth to well-being? Money matters, but less than people think. *Journal of Positive Psychology*, 4(6), 523–527.

- Aknin, L. B., Sandstrom, G. M., Dunn, E. W., & Norton, M. I. (2011). It's the recipient that counts: Spending money on strong social ties leads to greater happiness than spending on weak social ties. *PLoS One*, 6(2), e17018.
- Aknin, L. B., Dunn, E. W., Whillans, A. V., Grant, A. M., & Norton, M. I. (2013). Making a difference matters: Impact unlocks the emotional benefits of prosocial spending. *Journal of Economic Behavior and Organization*, 88, 90–95.
- Anik, L., & Norton, M. I. (2014). Matchmaking promotes happiness. *Social Psychological and Personality Science*, 5, 644–652.
- Baumeister, R. F., Vohs, K. D., Aaker, J. L., & Garbinsky, E. N. (2013). Some key differences between a happy life and a meaningful life. *Journal of Positive Psychology*, 8(6), 505–516.
- Bellezza, S., Gino, F., & Keinan, A. (2014). The red sneakers effect: Inferring status and competence from signals of nonconformity. *Journal of Consumer Research*, 41(1), 35–54.
- Berger, J., & Milkman, K. L. (2012). What makes online content viral? *Journal of Marketing Research*, 49(2), 192–205.
- Bhattacharjee, A., & Mogilner, C. (2014). Happiness from ordinary and extraordinary experiences. *Journal of Consumer Research*, 41(1), 1–17.
- Botti, S., & Hsee, C. K. (2010). Dazed and confused: How the temporal costs of choice freedom lead to undesirable outcomes. *Organizational Behavior and Human Decision Processes*, 112(2), 161–171.
- Botti, S., & McGill, A. L. (2006). When choosing is not deciding: The effect of perceived responsibility on satisfaction. *Journal of Consumer Research*, 33(2), 211–219.
- Buechel, E. C., & Janiszewski, C. (2014). A lot of work or a work of art: How the structure of a customized assembly task determines the utility derived from assembly effort. *Journal of Consumer Research*, 40(5), 960–972.
- Caprariello, P. A., & Reis, H. T. (2013). To do, to have, or to share? Valuing experiences over material possessions depends on the involvement of others. *Journal of Personality and Social Psychology*, 104(2), 199–215.
- Carstensen, L. L., Isaacowitz, D. M., & Charles, S. T. (1999). Taking time seriously: A theory of socioemotional selectivity. *American Psychologist*, 54(3), 165–181.
- Carter, T. J., & Gilovich, T. (2010). The relative relativity of material and experiential purchases. *Journal of Personality and Social Psychology*, 98(1), 146–159.
- Carter, T. J., & Gilovich, T. (2012). I am what I do, not what I have: The differential centrality of experiential and material purchases to the self. *Journal of Personality and Social Psychology*, 102(6), 1304–1317.
- Chan, C., & Mogilner, C. (2014). Experiential gifts are more socially connecting than material gifts. (Working paper.)
- Chan, E., & Mukhopadhyay, A. (2010). When choosing makes a good thing better: Temporal variations in the valuation of hedonic consumption. *Journal of Marketing Research*, 47(3), 497–507.
- Chernev, A. (2003). When more is less and less is more: The role of ideal point availability and assortment in consumer choice. *Journal of Consumer Research*, 30(2), 170–183.
- Cowley, E. (2008). The perils of hedonic editing. *Journal of Consumer Research*, 35(1), 71–84.
- DeCarlo, T. E., & Barone, M. J. (2009). With suspicious (but happy) minds: Mood's ability to neutralize the effects of suspicion on persuasion. *Journal of Consumer Psychology*, 19(3), 326–333.

- Deci, E. L., & Ryan, R. M. (1985). The general causality orientations scale: Self-determination in personality. *Journal of Research in Personality*, 19(2), 109–134.
- Diehl, K., & Poynor, C. (2010). Great expectations?! Assortment size, expectations, and satisfaction. *Journal of Marketing Research*, 47(2), 312–322.
- Diener, E., & Biswas-Diener, R. (2002). Will money increase subjective well-being? A literature review and guide to needed research. *Social Indicators Research*, 57(2), 119–169.
- Diener, E., Wirtz, D., Tov, W., Kim-Prieto, C., Choi, D., Oishi, S., & Biswas-Diener, R. (2010). New well-being measures: Short scales to assess flourishing and positive and negative feelings. *Social Indicators Research*, 97(2), 143–156.
- Dunn, E. W., Aknin, L. B., & Norton, M. I. (2008). Spending money on others promotes happiness. *Science*, 319(5870), 1687–1688.
- Dunn, E. W., Aknin, L. B., & Norton, M. I. (2014). Prosocial spending and happiness: Using money to benefit others pays off. *Current Directions in Psychological Science*, 23(1), 41–47.
- Dunn, E., & Norton, M. (2013). *Happy Money: The Science of Smarter Spending*. New York: Simon & Schuster.
- Dunn, E. W., Gilbert, D. T., & Wilson, T. D. (2011). If money doesn't make you happy, then you probably aren't spending it right. *Journal of Consumer Psychology*, 21(2), 115–125.
- Etkin, J. & Mogilner, C. (2015). Does variety increase happiness? (Working paper.)
- Franke, N., Schreier, M., & Kaiser, U. (2010). The “I designed it myself” effect in mass customization. *Management Science*, 56(1), 125–140.
- Frey, B. S., & Stutzer, A. (2000). Happiness, economy and institutions. *The Economic Journal*, 110(446), 918–938.
- Gilovich, T., Kumar, A., & Jampol, L. (in press). A wonderful life: Experiential consumption and the pursuit of happiness. *Journal of Consumer Psychology*.
- Gino, F., & Mogilner, C. (2014). Time, money, and morality. *Psychological Science*, 25(2), 414–421.
- Godes, D., & Mayzlin, D. (2004). Using on-line conversations to study word-of-mouth communication. *Marketing Science*, 23(4), 545–560.
- Godes, D., & Mayzlin, D. (2009). Firm-created word-of-mouth communication: Evidence from a field test. *Marketing Science*, 28(4), 721–739.
- Gourville, J. T., & Soman, D. (2005). Overchoice and assortment type: When and why variety backfires. *Marketing Science*, 24(3), 382–395.
- Gu, Y., Botti, S., & Faro, D. (2013). Turning the page: The impact of choice closure on satisfaction. *Journal of Consumer Research*, 40(2), 268–283.
- Haisley, E., Mostafa, R., & Loewenstein, G. (2008). Subjective relative income and lottery ticket purchases. *Journal of Behavioral Decision Making*, 21(3), 283–295.
- Hill, G., & Howell, R. T. (2014). Moderators and mediators of pro-social spending and well-being: The influence of values and psychological need satisfaction. *Personality and Individual Differences*, 69, 69–74.
- Howell, R. T., & Hill, G. (2009). The mediators of experiential purchases: Determining the impact of psychological needs satisfaction and social comparison. *The Journal of Positive Psychology*, 4(6), 511–522.
- Hsee, C. K., Yang, Y., Gu, Y., & Chen, J. (2009). Specification seeking: How product specifications influence consumer preferences. *Journal of Consumer Research*, 35(6), 952–966.

- Hsee, C. K., Yang, A. X., & Wang, L. (2010). Idleness aversion and the need for justified busyness. *Psychological Science*, 21(7), 926–930.
- Hsee, C. K., & Zhang J. (2004). Distinction bias: Misprediction and mischoice due to joint evaluation. *Journal of Personality and Social Psychology*, 86(5), 680–695.
- Iyengar, S. S., & Lepper, M. R. (2000). When choice is demotivating: Can one desire too much of a good thing? *Journal of Personality and Social Psychology*, 79(6), 346–357.
- Kahneman, D. (2011). *Thinking, Fast and Slow*. New York: Farrar, Straus and Giroux.
- Kahneman, D., & Deaton, A. (2010). High income improves evaluation of life but not emotional well-being. *Proceedings of the National Academy of Sciences*, 107(38), 16489–16493.
- Kahneman, D., Krueger, A. B., Schkade, D., Schwarz, N., & Stone, A. A. (2006). *Would you be happier if you were rich? A focusing illusion*. *Science*, 312(5782), 1908–1910.
- Keinan, A., & Kivetz, R. (2011). Productivity orientation and the consumption of collectable experiences. *Journal of Consumer Research*, 37(6), 935–950.
- Kramer, A. D., Guillory, J. E., & Hancock, J. T. (2014). Experimental evidence of massive-scale emotional contagion through social networks. *Proceedings of the National Academy of Sciences*, 111(24), 8788–8790.
- Kross, E., Verduyn, P., Demiralp, E., Park, J., Lee, D. S., Lin, N., & Shablack, H. (2013). Facebook use predicts declines in subjective well-being in young adults. *PLoS One*, 8(8), e69841.
- Kumar, A., Killingsworth, M. A., & Gilovich, T. (2014). Waiting for Merlot: Anticipatory consumption of experiential and material purchases. *Psychological Science*, 25(10) 1924–1931.
- Kuziemko, I., Buell, R. W., Reich, T., & Norton, M. I. (2014). Last-place aversion: Evidence and redistributive implications. *Quarterly Journal of Economics*, 129, 105–149.
- Labroo, A. A., & Mukhopadhyay, A. (2009). Lay theories of emotion transience and the search for happiness: A fresh perspective on affect regulation. *Journal of Consumer Research*, 36(2), 242–254.
- Labroo, A. A., & Patrick, V. M. (2009). Psychological distancing: Why happiness helps you see the big picture. *Journal of Consumer Research*, 35(5), 800–809.
- Langer, E. J. (1975). The illusion of control. *Journal of Personality and Social Psychology*, 32(2), 311–328.
- Langer, E. J., & Rodin, J. (1976). The effects of choice and enhanced personal responsibility for the aged: A field experiment in an institutional setting. *Journal of Personality and Social Psychology*, 34, 191–198.
- Larsen, J. T., & McKibban, A. R. (2008). Is happiness having what you want, wanting what you have, or both? *Psychological Science*, 19(4), 371–377.
- Lee, Y. H., & Qiu, C. (2009). When uncertainty brings pleasure: The role of prospect imageability and mental imagery. *Journal of Consumer Research*, 36(4), 624–633.
- Lehmann, D. R. (1998). Customer reactions to variety: Too much of a good thing? *Journal of the Academy of Marketing Science*, 26(1), 62–65.
- Liu, W., & Aaker, J. (2008). The happiness of giving: The time-ask effect. *Journal of Consumer Research*, 35(3), 543–557.
- Lyubomirsky, S., King, L., & Diener, E. (2005). The benefits of frequent positive affect: Does happiness lead to success? *Psychological Bulletin*, 131(6), 803–855.

- Meloy, M. G., Russo, E., & Miller, E. G. (2006). Monetary incentives and mood. *Journal of Marketing Research*, 43(2), 267–275.
- Mick, D. (2005, September). *The Presidential Address: Meaning and Mattering through Transformative Consumer Research*. Speech presented at the North American Conference of the Association for Consumer Research, San Antonio, TX.
- Mochon, D., Norton, M. I., & Ariely, D. (2008). Getting off the hedonic treadmill, one step at a time: The impact of regular religious practice and exercise on well-being. *Journal of Economic Psychology*, 29(5), 632–642.
- Mochon, D., Norton, M. I., & Ariely, D. (2012). Bolstering and restoring feelings of competence via the IKEA effect. *International Journal of Research in Marketing*, 29(4), 363–369.
- Mogilner, C. (2010). The pursuit of happiness: Time, money, and social connection. *Psychological Science*, 21(9), 1348–1354.
- Mogilner, C., & Aaker, J. (2009). “The time vs. money effect”: Shifting product attitudes and decisions through personal connection. *Journal of Consumer Research*, 36(2), 277–291.
- Mogilner, C., Aaker, J., & Kamvar, S. D. (2012). How happiness affects choice. *Journal of Consumer Research*, 39(2), 429–443.
- Mogilner, C., Chance, Z., & Norton, M. I. (2012). Giving time gives you time. *Psychological Science*, 23(10), 1233–1238.
- Mogilner, C., Kamvar, S. D., & Aaker, J. (2011). The shifting meaning of happiness. *Social Psychological and Personality Science*, 2, 395–402.
- Mogilner, C., & Norton, M. (2015). Choice of happiness: Experienced versus remembered. (Working paper.)
- Mogilner, C., Rudnick, T., & Iyengar, S. S. (2008). The mere categorization effect: How the presence of categories increases choosers’ perceptions of assortment variety and outcome satisfaction. *Journal of Consumer Research*, 35(2), 202–215.
- Mogilner, C., Shiv, B., & Iyengar, S. S. (2013). Eternal quest for the best: Sequential (vs. simultaneous) option presentation undermines choice commitment. *Journal of Consumer Research*, 39(6), 1300–1312.
- Moreau, C. P., Bonney, L., & Herd, K. B. (2011). It’s the thought (and the effort) that counts: How customizing for others differs from customizing for oneself. *Journal of Marketing*, 75(5), 120–133.
- Nicolao, L., Irwin, J. R., & Goodman, J. K. (2009). Happiness for sale: Do experiential purchases make consumers happier than material purchases? *Journal of Consumer Research*, 36(2), 188–198.
- Norton, M. I., Anik, L., Aknin, L. B., & Dunn, E. W. (2011). Is life nasty, brutish, and short? Philosophies of life and well-being. *Social Psychological and Personality Science*, 2(6), 570–575.
- Norton, M. I., Frost, J. H., & Ariely, D. (2007). Less is more: The lure of ambiguity, or why familiarity breeds contempt. *Journal of Personality and Social Psychology*, 92(1), 97–105.
- Norton, M. I., Mochon, D., & Ariely, D. (2012). The IKEA effect: When labor leads to love. *Journal of Consumer Psychology*, 22(3), 453–460.
- Ordabayeva, N., & Chandon, P. (2011). Getting ahead of the Joneses: When equality increases conspicuous consumption among bottom-tier consumers. *Journal of Consumer Research*, 38(1), 27–41.

- Poynor, C., & Wood, S. (2010). Smart subcategories: How assortment formats influence consumer learning and satisfaction. *Journal of Consumer Research*, 37(1), 159–175.
- Qiu, C., & Yeung, C. W. (2008). Mood and comparative judgment: Does mood influence everything and finally nothing? *Journal of Consumer Research*, 34(5), 657–669.
- Quoidbach, J., Gruber, J., Mikolajczak, M., Kogan, A., Kotsou, I. & Norton, M. I. (2014). Emodiversity and the emotional ecosystem. *Journal of Experimental Psychology: General*, 143, 2057–2066.
- Raghnathan, R., & Corfman, K. (2006). Is happiness shared doubled and sadness shared halved? Social influence on enjoyment of hedonic experiences. *Journal of Marketing Research*, 43(3), 386–394.
- Ramanathan, S., & McGill, A. L. (2007). Consuming with others: Social influences on moment-to-moment and retrospective evaluations of an experience. *Journal of Consumer Research*, 34(4), 506–524.
- Redden, J. P. (2008). Reducing satiation: The role of categorization level. *Journal of Consumer Research*, 34(5), 624–634.
- Rosenzweig, E., & Gilovich, T. (2012). Buyer's remorse or missed opportunity? Differential regrets for material and experiential purchases. *Journal of Personality and Social Psychology*, 102(2), 215–223.
- Rucker, D. D., Dubois, D., & Galinsky, A. D. (2011). Generous paupers and stingy princes: Power drives consumer spending on self versus others. *Journal of Consumer Research*, 37(6), 1015–1029.
- Rudd, M., Vohs, K., & Aaker, J. (2012). Awe expands people's perception of time, alters decision making and enhances well-being. *Psychological Science*, 23(10), 1130–1136.
- Ryan, R. M., & Deci, E. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68–78.
- Seligman, M. E. P. (1998, September). *The President's Address (Annual Report)*. Speech presented at the National Press Club, Washington, DC.
- Sellier, A. L., & Avnet, T. (2014). So what if the clock strikes? Scheduling style, control, and well-being. *Journal of Personality and Social Psychology: Attitudes and Social Cognition*, 107(5), 791–808.
- Shah, A. K., & Alter, A. L. (2014). Consuming experiential categories. *Journal of Consumer Research*, 41(4).
- Simonson, I. (1990). The effect of purchase quantity and timing on variety seeking behavior. *Journal of Marketing Research*, 27(2), 150–162.
- Su, R., Tay, L., & Diener, E. (2014). The development and validation of Comprehensive Inventory of Thriving (CIT) and Brief Inventory of Thriving (BIT). *Applied Psychology: Health and Well-Being*.
- Van Boven, L., & Gilovich, T. (2003). To do or to have? That is the question. *Journal of Personality and Social Psychology*, 85(6), 1193–1202.
- Vohs, K. D., Wang, Y., Gino, F., & Norton, M. I. (2013). Rituals enhance consumption. *Psychological Science*, 24(9), 1714–1721.
- Whitchurch, E. R., Wilson, T. D., & Gilbert, D. T. (2010). "He loves me, he loves me not . . .": Uncertainty can increase romantic attraction. *Psychological Science*, 22, 172–175.

- Wilkinson, R., & Pickett, K. (2010). *The Spirit Level: Why Equality Is Better for Everyone*. London: Penguin.
- Wilson, T. D., Centerbar, D. B., Kermer, D. A., & Gilbert, D. T. (2005). The pleasures of uncertainty: Prolonging positive moods in ways people do not anticipate. *Journal of Personality and Social Psychology*, 88(1), 5–21.
- Yang, A. X., Hsee, C. K., Liu, Y., & Zhang, L. (2011). The supremacy of singular subjectivity: Improving decision quality by removing objective specifications and comparisons. *Journal of Consumer Psychology*, 21(4), 393–404.
- Zauberman, G., Ratner, R. K., & Kim, K. (2009). Memories as assets: Strategic memory protection in choice over time. *Journal of Consumer Research*, 35(5), 715–728.

2 Attitude Change and Persuasion

Past, Present, and Future Directions

Zakary L. Tormala and Pablo Briñol

A fundamental goal of consumer psychology research is to shed light on the underlying psychological factors that drive consumer behavior. With this objective in mind, consumer psychologists have long emphasized the importance of understanding *attitudes*. Attitudes refer to the general and relatively enduring evaluations people have of other people, objects, or ideas. Stated differently, attitudes represent the extent to which one likes or dislikes something – for example, a product, company, or brand. Because attitudes can be one of the core drivers of consumer behavior (e.g., choosing one product over another, purchasing now versus later, spending more versus less, and so on), they have been the subject of considerable scrutiny in consumer psychology. Not surprisingly, perhaps, consumer psychologists have been especially interested in understanding the means by which consumers' attitudes can be shaped or changed, particularly in the context of persuasive messages.

In this chapter, we provide an overview of past, present, and future research on attitude change and persuasion. More specifically, we (1) review some of the classic research on persuasion to provide an overview of this area, (2) highlight what we view as some of the crucial recent developments in this field, and then (3) discuss several unanswered questions and opportunities for future research. Our goal is not to be exhaustive in our coverage of any of one of these topics, but rather to offer an illustrative review of the field and, hopefully, some encouragement to pursue what we see as some of the novel and important next steps. Before turning to our review, it is useful to consider some general background issues.

Attitudes

As noted, attitudes refer to our general evaluations of people (e.g., politicians), objects (e.g., new products), or issues (e.g., social policies). Attitudes can vary in a number of important ways. First, of course, they can vary in *valence*. Some attitudes are positive, some are negative, and others are relatively neutral. Consumers' attitudes toward a particular car, for instance, might range from positive (liking it) to neutral (neither liking nor disliking it) to negative (disliking it). Moreover, attitudes can differ in their *extremity*, or the extent to which they deviate from neutral (Abelson, 1995). Indeed, two

consumers could both report liking a car (i.e., have attitudes of the same valence), but differ in the extent to which they like it, such that one has a mild liking for the car and the other a more extreme liking. Given the well-documented link between attitudes and behavior (e.g., Fishbein & Ajzen, 1975), these individuals could be expected to behave differently toward the car despite both holding positive attitudes.

In addition to varying in valence and extremity, attitudes can differ in their underlying *bases*. According to the classic tripartite theory (Rosenberg & Hovland, 1960; Zanna & Rempel, 1988), attitudes can be based to varying degrees on affect or feelings (e.g., “this car feels fun to drive”), cognition or beliefs (e.g., “this car is fuel efficient”), and past or current behavior (e.g., “we have always driven this kind of car in my family”). Interestingly, people’s perceptions of their attitude bases can be as important as their actual attitude bases in determining their future thoughts and actions (e.g., See, Petty, & Fabrigar, 2008). Generally speaking, persuasive messages are more effective at inducing attitude change when they match (e.g., an emotional appeal targeting an affect-based attitude) rather than mismatch (e.g., a rational appeal targeting an affect-based attitude) the actual or perceived basis of the target attitude (e.g., Edwards, 1990; Fabrigar & Petty, 1999; Mayer & Tormala, 2010; See, Petty, & Fabrigar, 2008; See, Fabrigar, & Petty, 2013). As noted, people can base their attitudes not only on affect and cognition, but also on behavioral information (see Briñol & Petty, 2008). We will address this embodiment approach in the present review by describing research showing that people’s physical postures, facial expressions, and bodily movements can all play an important role in attitude formation and change.

Attitudes can also differ in their underlying strength. *Attitude strength* refers to the durability and impact of an attitude (Petty & Krosnick, 1995). Durability encapsulates an attitude’s resistance to change, such that strong (relative to weak) attitudes generally persist longer over time and are more resistant in the face of attack. Impact refers to the attitude’s influence over thoughts and behavior; in general, strong attitudes exert greater influence on thoughts (e.g., produce more attitude-consistent thinking) and behavior (e.g., produce more attitude-consistent behavior) than do their weaker counterparts. Thus, attitude strength can be a crucial moderator of persuasion effects – making it harder to change attitudes in some cases than in others – and also a crucial target of influence (e.g., strategically increasing consumers’ attitude strength when they already hold the desired position).

Over the years, attitude strength has been construed, and operationally defined, in a number of different ways (see Bassili, 1996; Petty & Krosnick, 1995). Some indicators of strength have been viewed as relatively objective in nature. For instance, attitude extremity (distance from neutrality; Abelson, 1995), attitude accessibility (the speed with which an attitude comes to mind; Fazio, 1995), and attitude ambivalence (the presence of conflicting positive and negative evaluations of the same object, which tends to weaken an attitude; Thompson, Zanna, & Griffin, 1995) can all be viewed as somewhat objective in

that they do not require people to introspect and report their perception of their own strength. Indeed, people can be ambivalent and not even know it if, for example, they have just a single consciously endorsed attitude that is univalent (e.g., “I do not like scary movies at all”) but contradicted by opposing cultural associations (“scary movies are quite popular”) or even a former attitude that is now rejected (“I used to like them”). In such cases, people can show signs of *implicit ambivalence* that weakens their attitudes even though they do not perceive or endorse having any attitude conflict (Petty & Briñol, 2012; Petty, Briñol, & DeMarree, 2007; Petty, Tormala, Briñol, & Jarvis, 2006).

Other strength indicators are more subjective in nature in that they revolve around people’s perceptions of their own attitudes and whether they are personally important (Boninger, Krosnick, Berent, & Fabrigar, 1995), backed by extensive knowledge (Davidson, Yantis, Norwood, & Montano, 1985), and held with confidence or certainty (for reviews, see Rucker, Tormala, Petty, & Briñol, 2014; Tormala & Rucker, 2007). Of importance, regardless of the particular operationalization of strength in a study, and whether it is viewed as objective or subjective, the dominant perspective is that each dimension or type of strength matters because it shapes the attitude’s durability and impact.

Finally, attitudes themselves, and the measures used to study them, have been viewed as varying along a continuum ranging from explicit to implicit. Indeed, after a long tradition of assessing the impact of influence treatments, such as persuasive messages, on attitudes using deliberative self-report measures (Eagly & Chaiken, 1993; Petty & Wegener, 1998), recent work has also used measures that tap the more automatic evaluations associated with objects, issues, and people. Techniques that assess automatic evaluative associations without directly asking people to report their attitudes are often referred to as *implicit measures*, whereas assessments that tap more deliberative and acknowledged evaluations are referred to as *explicit measures* (see Gawronski & Payne, 2010; Petty, Fazio, & Briñol, 2009, for reviews). Although implicit and explicit measures sometimes appear discrepant (e.g., Briñol, Petty, & Wheeler, 2006), they do often yield the same outcome (e.g., both revealing that a person likes a particular brand). In the present review, we will focus mainly on deliberative, self-report attitudes as those have been the primary focus of researchers in consumer psychology, though recent research has begun to test the effects of persuasive messages on implicit attitudes as well (Briñol, Petty, & McCaslin, 2009; Perkins & Forehand, 2010; Smith, De Houwer, & Nosek, 2013).

Persuasion: Source, Message, and Recipient Factors

There are many ways to organize the literature on attitude change and persuasion. One way, which we follow in the current chapter, revolves around the basic distinction among source, message, and recipient variables. Stated differently, researchers often think about persuasion in terms of *who* says *what*

to *whom*. Indeed, classic persuasion research essentially viewed these categories as the “3 Ws” of persuasion, whereby maximizing relevant variables within these buckets would facilitate successful persuasion outcomes. In the 1950s, when persuasion became a central focus of social and consumer psychology research, this was the prevailing view on the factors determining a given message’s likely success or failure. Since then, we have learned much more about the underlying psychological mechanisms driving persuasion and the interactions among source, message, and recipient factors, but it remains a useful trichotomy for organizing the literature. For the remainder of this chapter, we will briefly review some of the classic research on persuasion, dive deeper into recent developments in this field, and then highlight open questions and new directions, using the source-message-recipient framework as a guide for organizing and understanding the literature.

Classic Conceptualizations: Who Says What to Whom?

Source Factors

Source factors in persuasion refer to aspects of the individual or company that crafts and delivers the persuasive message. There is a voluminous body of research examining the many dimensions of message sources that can play a role in persuasion (for a review, see Briñol & Petty, 2009a). These dimensions include source credibility, attractiveness, and numerical status, among many others. The classic view on these source dimensions is that because they are positively valenced, more of them – for example, more credibility or more attractiveness – fosters more persuasion. Thus, the conventional persuasive prescription was to increase favorable source factors as a means to persuasion. For a historical review on this main effect approach to persuasion, we refer readers to Petty and Briñol (2008).

Perhaps the most frequently studied source factor is source credibility (Kelman & Hovland, 1953; for reviews see Petty & Wegener, 1998; Pornpitakpan, 2004). Source credibility is an umbrella construct that captures both the expertise and trustworthiness of the source of a message. *Expertise* generally refers to the amount of knowledge one has or one’s perceived ability to provide accurate and truthful information. *Trustworthiness* describes one’s perceived honesty or motivation to provide accurate and truthful information. In general, whether credibility is operationalized in terms of expertise (Rhine & Severance, 1970) or trustworthiness (Mills & Jellison, 1967), the classic finding is that the more credibility the source of a message has, the more persuaded his or her audience will be. As noted by Petty and colleagues (e.g., Petty, 1997; Petty & Briñol, 2008), early theories of persuasion often suggested that there was likely to be just one mechanism or process that was responsible for whichever outcome was produced. For example, an expert source was thought to increase persuasion by getting a person to learn or internalize the message arguments (Kelman, 1958).

In addition to credibility, other source dimensions such as attractiveness have proven important to persuasion. In general, the more the audience likes the source, the more persuasion there tends to be. As one example, physically attractive sources have been found to be more persuasive than physically unattractive sources (Snyder & Rothbart, 1971). Relatedly, a source's celebrity status has been shown to influence persuasion (e.g., Petty, Cacioppo, & Schumann, 1983); under some conditions, celebrities are more persuasive than unfamiliar sources. Furthermore, consistent with the widely documented similarity-attraction effect (i.e., the notion that people tend to like similar others more than dissimilar others; Byrne, 1971), more persuasion tends to occur when the source of a message is similar to the audience. And importantly, source similarity can increase persuasion regardless of whether the similar dimension is important (e.g., when the source and recipient share important values) or unimportant (e.g., when the source simply mimics the recipient's subtle non-verbal behaviors; Fleming & Petty, 2000). Importantly, while source credibility has been argued to induce agreement because of increased acceptance of the message arguments, the traditional view of attractiveness was that it increases persuasion by boosting identification with the message source (Kelman, 1958).

Potentially related to the dimensions of credibility and attractiveness, a source's numerical status – that is, the degree to which the source is in the statistical majority or minority on a topic – can play an important role in persuasion. All else being equal, people have been shown to be more influenced by numerical majorities than by numerical minorities (see Wood et al. 1994; Tormala, Petty, & DeSensi, 2010). For instance, a persuasive message that ostensibly represents 75 percent of surveyed consumers' views tends to be more effective at inducing attitude change than a message representing only 25 percent of consumers' views. Traditionally, majorities were thought to be more persuasive because people seek to belong to and be accepted by the majority group (Moscovici, 1980, 1985).

Finally, source power can be important as well. Power typically involves an individual's perceived ability to control others' outcomes by providing or withholding rewards or punishments (see Chapter 12 in this volume). In general, powerful sources have been found to produce more attitude change and persuasion than powerless sources (e.g., Festinger & Thibaut, 1951; French & Raven, 1959). Classic work on this topic suggested that this effect stems from mere compliance and should be attenuated in the absence of the powerful source (e.g., Kelman, 1958).

Message Factors

In addition to source factors, message factors are crucial to persuasion. Message factors refer to characteristics of the message itself; for example, what is written or spoken in an advertisement, product review, or restaurant recommendation. Viewed differently, whereas source factors refer to features of the person or company delivering the message, message factors pertain to what that person or

company actually says. Broadly speaking, message factors can be viewed in terms of what the source says (e.g., what position they take, how compelling the arguments are), how much of it they say (e.g., number of arguments), and how often they say it (e.g., argument repetition).

Perhaps the most fundamental aspect of a persuasive message is its position, or direction. That is, does the message argue against or in favor of the attitude object? Even more crucial in some ways is whether the message is proattitudinal or counterattitudinal, meaning it is congruent or incongruent with recipients' initial attitudes, respectively (Clark & Wegener, 2013). All else equal, proattitudinal messages tend to be more convincing and persuasive than counterattitudinal messages (e.g., Lord, Ross, & Lepper, 1979). The traditional explanation for this effect is that proattitudinal messages fall within the recipient's latitude of acceptance (Sherif & Hovland, 1961) and, thus, are evaluated more favorably.

A related distinction is whether a message is one-sided or two-sided (Hovland, Lumsdaine, & Sheffield, 1949). Whereas one-sided messages contain arguments that advocate exclusively in the direction of the advocated position or product, two-sided messages contain arguments both against and in favor of it. A considerable amount of research has shown that two-sided messages can be more persuasive than one-sided messages, and this has been thought to occur because considering both sides increases the source's perceived trustworthiness (Crowley & Hoyer, 1994; Etgar & Goodwin, 1982; Pechmann, 1992). Even simply acknowledging minor blemishes or product flaws has been found to boost persuasion under some conditions by making the remaining positive features seem even more positive (Ein-Gar, Shiv, & Tormala, 2012).

Beyond the direction or valence of arguments in a message, there are many other message features that play an important role in persuasion. For example, argument quality – or message strength – refers to how compelling or convincing the arguments in a message are. For instance, a strong message that advocates changing a particular behavior (such as exercising) might highlight consequences of that behavior that are highly desirable, highly likely, and extremely important. A weaker message, by contrast, would highlight consequences that are less desirable, likely, or important. A voluminous body of research suggests that people tend to be more persuaded by strong rather than weak arguments (see Petty & Cacioppo, 1986; Petty & Wegener, 1998), because strong arguments elicit more favorable message-relevant thoughts.

The sheer number of arguments presented in a message can impact persuasion as well. Specifically, messages with more arguments often lead to more persuasion than messages with fewer arguments (Calder, Insko, & Yandell, 1974; Josephs, Giesler, & Silvera, 1994; Petty & Cacioppo, 1984). Similarly, in product evaluation studies, it has been shown that products with more attributes sometimes appear more desirable to consumers than products with fewer attributes (e.g., Sela & Berger, 2012). This effect is consistent with the notion of a general numerosity heuristic (Pelham, Sumarta, & Myaskovsky, 1994), in which people operate according to a “more is better” rule.

As a final example, research has found that simply repeating a message can increase persuasion. This effect was originally explained by the fact that repetition facilitates learning the message arguments. As it turned out, the number of repetitions and other contextual factors were critical to this effect. For example, a few repetitions enhances persuasion, but many repetitions can undo and even reverse this effect (Cacioppo & Petty, 1979; 1989; Gorn & Goldberg, 1980; Miller, 1976). The logic is similar to that articulated in the literature on mere repeated exposure (e.g., Bornstein, 1989; Grush, 1976; Zajonc, 1968), where initial repetition enhances positive associations but over-exposure can attenuate this effect.

Recipient Factors

Finally, recipient factors also are a critical determinant of persuasion. These refer to features or characteristics of the target audience – that is, the individuals who will receive the persuasive message. Recipient factors can be stable or situational (e.g., intelligence or current mood, respectively) and can vary along numerous dimensions. Of central import in early persuasion research were variables that affect motivation and the ability to process and understand persuasive messages, as well as variables that directly affect interpretations of those messages. Consider intelligence. Past research shows that as an audience's intelligence increases, the audience becomes more difficult to persuade (Rhodes & Wood, 1992). The initial explanation for this effect was that intelligence provides individuals with the ability both to evaluate the persuasiveness of an appeal and to counter the information presented.

Of course, there are numerous recipient factors relevant to persuasion beyond intelligence, such as self-esteem, personality, knowledge, and many others. Perhaps the most studied recipient variable concerns the emotional state of the recipient of the persuasion attempt (Petty & Briñol, 2015). As one classic example, happiness has been shown to increase persuasion as a result of classical conditioning (i.e., people associating their current happiness with the message or attitude object). Another prominent recipient factor that has received considerable attention pertains to the recipient's overt physical behavior, or what they are currently doing with their bodies (see Briñol & Petty, 2008). For instance, Cacioppo, Priester, and Berntson (1993) found that neutral Chinese ideographs (irrelevant stimuli for the sample of participants) presented during arm flexion (a pulling or approach behavior) were subsequently evaluated more favorably than ideographs presented during arm extension (a pushing or avoidance behavior). Similarly, Wells and Petty (1980) found that inducing mere head nodding could influence one's agreement with a persuasive message. Like mood effects, one reason these embodiment effects could occur is through classical conditioning, whereby positive behaviors become directly associated with the attitude object (Staats & Staats, 1958).

Another recipient factor, which we devote more attention to later in this chapter, includes recipients' "cognitive feelings" (Clore & Parrott, 1994), such

as processing ease or fluency. In general, ease is associated with more persuasion than difficulty; that is, people form more favorable attitudes toward stimuli that feel easy to process (e.g., Labroo, Dhar, & Schwarz, 2008; Lee & Labroo, 2004) and are more persuaded when they can easily generate supportive reasons (e.g., because just a few are requested; Wänke, Böhner, & Jurkowitsch, 1997). The original account for this effect made reference to an availability heuristic whereby people infer that there were more reasons available when it was easy rather than difficult to generate examples (Schwarz et al., 1991).

Contextual Factors: When and Why

After a few decades of research focusing on understanding source, message, and recipient factors as a prescription for successful persuasion, researchers in the 1980s turned their attention away from asking *whether* these variables affected persuasion and instead began to emphasize *when* and *why* they affect persuasion (for a more thorough discussion of the reasons for this shift, see Petty, 1997). This new wave of research provided deeper insight into a number of puzzling issues in the field, including why some factors (e.g., an expert source) sometimes increased persuasion but in other cases undermined it, as well as why attitude change sometimes seemed durable and impactful (e.g., producing attitudes that guide behavior) but other times seemed rather transitory or ephemeral.

The result of this shifted focus was the development of new theories capable of accounting for multiple effects, processes, and consequences of persuasion. Of particular import were the elaboration likelihood model (ELM; Petty & Briñol, 2012; Petty & Cacioppo, 1986), and the heuristic-systematic model (HSM; Chaiken & Ledgerwood, 2012; Chaiken, Liberman, & Eagly, 1989). These frameworks helped shed new light on the multiple ways a given variable could affect attitude change in different contexts. One crucial insight, first articulated in the ELM, was that the psychological processes mediating the effects of any persuasion variable on attitude change could be organized into a finite set that operate at different points on an elaboration continuum. That is, the process guiding a given variable's effect on persuasion – determining the presence or absence of the effect, the direction of the effect, and the durability and downstream impact of the effect – depended critically on the amount of thinking the recipient was likely to engage in. Under low thinking conditions, variables can affect persuasion by operating as a simple cue or heuristic. Under relatively high thinking conditions, the same variables might impact persuasion through more thoughtful means such as by affecting the direction of thoughts that come to mind or serving as a piece of evidence (i.e., an argument) to be scrutinized. When thinking is not constrained to be low or high, the same variables can shape persuasion by affecting the amount of message-relevant thinking that occurs. In the parlance of the ELM, the processes engaged at the low end of the elaboration continuum are referred to as the *peripheral route*,

whereas processes engaged at the high end of the continuum are referred to as the *central route*. The more persuasion is based on thoughtful processing – that is, the more central route the process is – the more it tends to persist over time, resist attempts at change, and have consequences for other judgments and behavior (Petty, Haugtvedt, & Smith, 1995).

To offer one example of the organizational and predictive utility of this multiprocess framework, consider source credibility. Traditional views held that credible sources were more persuasive because people have a heuristic that, “if an expert says it, it must be true” (Chaiken, 1980). Consistent with this perspective, early research revealed that source credibility effects on persuasion are more prominent when people are not very motivated or able to think carefully (Petty, Cacioppo, & Goldman, 1981). However, subsequent studies suggested that source credibility does not always operate according to a simple cue or heuristic. Also, more credibility does not always foster more persuasion. That may be the dominant effect, but a more nuanced look indicates that source credibility can produce different effects in different circumstances.

For example, when thinking is not initially constrained to be high or low, credibility can influence persuasion by affecting the amount of thinking people do. When people are unsure whether a message warrants or needs scrutiny, they can use the credibility of the source as a guide – for instance, thinking more about a message from a knowledgeable source than a source lacking in knowledge (e.g., Heesacker, Petty, & Cacioppo, 1983). Furthermore, if high credibility entices someone to think more about a message but that message has weak arguments, then high credibility can be associated with *reduced* persuasion, the opposite of its effect when serving as a simple heuristic. Consistent with this logic, having an expert source sometimes increases persuasion when message arguments are strong but decreases it when message arguments are weak (Bohner, Ruder, & Erb, 2002; Heesacker, Petty, & Cacioppo, 1983).

When processing motivation and ability are high, such as when the message feels highly relevant (Petty & Cacioppo, 1979) and recipients are undistracted (Petty, Wells, & Brock, 1976), people think more carefully about a message. Nevertheless, that thinking can be quite biased – for example, by source credibility. Chaiken and Maheswaran (1994) found that when a message topic was personally important but ambiguous, meaning that recipients were motivated to think about the message but it was not clearly cogent or specious in its own right, an expert (vs. non-expert) source produced more persuasion by biasing the direction of people’s thoughts. Under very similar conditions, Tormala, Briñol, and Petty (2006) also found that credible sources could increase persuasion by biasing the valence of recipients’ message-relevant thinking.

By now, considerable research has demonstrated that numerous variables beyond source credibility can have this same complexity of effects and mechanisms depending on recipients’ initial motivation to process (see Petty & Wegener, 1998). As one additional example, a message recipient’s happiness can act as a simple cue to persuasion under low thinking conditions (Slovic, Finucane, Peters, & Macgregor, 2002), be analyzed as an argument (Martin, Ward,

Achee, & Wyer, 1993) or bias the direction of thoughts (Petty, Schumann, Richman, & Strathman, 1993) under high thinking conditions, and influence his or her amount of thinking when that thinking is initially unconstrained to be high or low (Schwarz, Bless, & Bohner, 1991). Based on its ability to integrate this diversity of effects and processes, the ELM emerged as the dominant theory of persuasion, specifying the processes by which source, message, recipient, and context factors shape persuasive outcomes and processes (for historical reviews of the ELM, see Briñol & Petty, 2012; Petty & Briñol, 2012).

Current Developments: A Metacognitive Reconceptualization

As just described, following the first few decades of research on persuasion, we gained much insight into the complexity of source, message, and recipient effects with the development of multiprocess theories such as the ELM and HSM. Research guided by these frameworks shed light on the multiple roles that source, message, and recipient variables play in persuasion. Continuing in this vein, research in the past two decades years has made great strides in deepening our understanding of the processes through which persuasion variables exert their impact and expanding our insight into their diverse and previously unidentified outcomes. Perhaps the greatest shift in this domain has been the new focus on metacognitive factors in persuasion. *Metacognition* refers to thinking about thinking – that is, people’s thoughts about their own or other people’s thoughts, judgments, and mental processes (for a review, see Briñol & DeMarree, 2012; Petty, Briñol, Tormala, & Wegener, 2007). Recently, there has been an enormous influx of research investigating metacognitive inputs, mechanisms, and outcomes in persuasion.

One of the most influential demonstrations that people’s thoughts about their thoughts can be consequential came from research on the *ease of retrieval* paradigm. In the original study on this topic, Schwarz, Bless, Strack, and colleagues (1991) asked participants to list either six (which was easy) or twelve (which was difficult) examples of their own assertiveness. Interestingly, people asked to retrieve fewer examples subsequently viewed themselves as *more* assertive, because their experience of generating or retrieving relevant cognitions was subjectively easier. This cognitive ease, in turn, can lead people to infer that many more such cognitions are likely to be available (Schwarz, 2004) or that those cognitions that came to mind are valid and can be held with confidence (Tormala, Petty, & Briñol, 2002). As we will discuss, this basic idea and finding have fostered numerous and important insights in attitude change and persuasion.

Rucker and Tormala (2012) recently reviewed other metacognitive approaches relevant to the consumer domain. As just as one example, the persuasion knowledge model (Friestad & Wright, 1994, 1995; Kirmani & Campbell, 2004) proposes that consumers hold lay theories about persuasion involving the agent of persuasion, the target of persuasion, and their

interaction. Moreover, consumers use these theories to evaluate persuasion attempts (e.g., advertisements and sales pitches), assess their validity and sincerity, and determine whether it is appropriate or inappropriate to be influenced by them. These perceptions, or lay theories, are then used as a basis for responding and yielding to or resisting the influence attempt (for a recent empirical example of naive theories about consumer persuasion, see Briñol, Rucker, & Petty, 2015). Other examples of metacognitive approaches in consumer research that are relevant to understanding attitude and persuasion processes include the accessibility-diagnostics model (Feldman & Lynch, 1988; Lynch, 2006) and the multiple pathway anchoring and adjustment model (Cohen & Reed, 2006).

In this new wave of metacognition research, particular emphasis has been placed on studying the sense of certainty people have about their own thoughts and attitudes. In general, being certain about a thought or attitude means that one holds the thought or attitude with confidence or conviction and believes it is valid (Rucker et al., 2014). This feeling of confidence or conviction can stem from one's perception that the thought or attitude is correct or simply that it is clear in one's mind (Petrocelli, Tormala, & Rucker, 2007). In either case, certainty can have crucial implications for persuasion and other attitude-relevant outcomes.

For the most part, recent insights into the role of psychological certainty in persuasion has emanated from two research streams. One, termed the self-validation hypothesis (Briñol, Petty, & Tormala, 2004; Petty, Briñol, & Tormala, 2002), has focused on the role of *thought confidence* in persuasion. Conventional research in persuasion largely focused on the amount and valence of people's thinking when they receive persuasive messages. In addition to these two dimensions of thought, the self-validation hypothesis introduced the notion of thought confidence. In essence, the notion is that when individuals receive persuasive messages, it is not only the number and direction of their thoughts that matter but also the confidence with which they hold those individual thoughts (and the extent to which they like their thoughts). Thoughts that are liked and held with high confidence exert more impact on persuasion than thoughts that are disliked or held with low confidence.

In fact, some studies have revealed that having more thoughts in one direction (e.g., positive) can push attitudes in the other direction (e.g., negative) if those thoughts are held with low confidence (e.g., Tormala, Petty & Briñol, 2002). Similarly, research has demonstrated that increasing positive dimensions of a given persuasion variable (e.g., using a source with greater credibility) can sometimes undermine persuasion if doing so boosts confidence in negative thoughts (e.g., because the message contains weak arguments; Tormala, Briñol, & Petty, 2006). These effects are particularly likely to emerge if people are engaged in thoughtful processing and have both the motivation and ability to reflect on their own thinking. Thus, like the ELM, the self-validation hypothesis identifies conditions under which, and new and unique processes through which, normally positive factors can have negative persuasion effects and vice versa.

Another metacognitive factor that has received considerable scrutiny in recent persuasion research is *attitude certainty*. Although attitude certainty has a storied history in attitudes research (see Gross, Holtz, & Miller, 1995), its emergence as a central variable in persuasion is a more recent development (see Rucker et al., 2014). This emergence began with research on resistance to persuasion (Tormala, Clarkson, & Petty, 2006; Tormala & Petty, 2002; see also Petty, Tormala, & Rucker, 2004; Tormala, 2008), where it was observed that, contrary to previous assumptions, when people resisted persuasive attacks on their attitudes, those attitudes did not remain literally unchanged. On the contrary, when people resist persuasion (meaning they received a persuasive message, but their attitude valence and extremity were unchanged), they can perceive and reflect upon this resistance and then make upward or downward adjustments to attitude certainty. Consequently, persuasive messages that seem on the surface to have had no effect on attitude change can actually have an important impact on attitude certainty, which has implications for future behavior, future openness to persuasion, and so on.

Based on these kinds of findings, a great deal of persuasion research has turned from the traditional focus on the amount and direction of thoughts, or even on attitudes per se, to understanding changes in the certainty people associate with those thoughts and attitudes. This metacognitive shift has expanded existing understandings of the effects of persuasion variables on attitudinal outcomes, and it has deepened our insight into the mechanisms driving those effects. In this section, we revisit the source-message-recipient trichotomy to highlight some of the novel insights we have gained as a result of this new metacognitive emphasis. In some areas (e.g., source factors), there has been considerable research on both thought confidence and attitude certainty. In other areas, the attention has been relatively more focused on either attitude certainty (e.g., message factors) or thought confidence (e.g., recipient factors). Thus, it is a developing area of research with much work to be done. In this section, we review relevant work on psychological certainty following the source-message-recipient trichotomy.

Source Effects on Certainty

Consider source effects on metacognitive certainty. These effects have been examined in the context of both thought confidence (and the self-validation hypothesis) and attitude certainty, but in both domains the common emphasis has been on understanding the ways in which classic source variables in persuasion affect metacognitive certainty.

First, various source factors have been shown to influence people's perceptions of their own thoughts. For example, in addition to providing a simple heuristic to persuasion or influencing the amount or valence of thoughts people generate while processing a message, source credibility is now known to influence the confidence with which people hold their thoughts that come to mind during persuasive messages. Consistent with the logic that credibility influences

the perceived validity of information contained in a message (Kaufman, Stasson, & Hart, 1999), people tend to have more confidence in their message-relevant thoughts when source expertise and trustworthiness are high rather than low (e.g., Briñol, Petty, & Tormala, 2004). Moreover, because of this effect, source credibility can under some conditions backfire and undermine persuasion. Specifically, when people receive a message containing weak arguments and have negative thoughts about it, high source credibility can increase confidence in those negative thoughts (Tormala, Briñol, & Petty, 2006) and undermine persuasion. Thus, the metacognitive perspective allows us to predict and understand further instances in which normally positive persuasion variables have negative implications for persuasion.

Importantly, this metacognitive effect of source credibility depends on two critical moderators. First, source credibility affects persuasion through thought confidence only under high thinking conditions, such as when participants are relatively high rather than low in need for cognition (Cacioppo & Petty, 1982). Indeed, considering the validity of one's own thoughts requires both motivation and ability to process (Petty, Briñol, & DeMarree, 2007). Under low thinking conditions, source credibility operates as a simple cue to persuasion, as in past research (Petty, Cacioppo, & Goldman, 1981). Second, Tormala, Briñol, and Petty (2007) demonstrated that source credibility affects thought confidence only when the source information follows rather than precedes the persuasive message. When source information precedes a message under high thinking conditions, it biases the direction or valence of thoughts, as reviewed earlier (Chaiken & Maheswaran, 1994). In short, elaboration and timing are two important moderators of self-validation effects.

Like source credibility, majority versus minority source status can affect persuasion by influencing the confidence with which people hold their individual thoughts. In one study, Horcajo, Petty, and Briñol (2010) presented participants with a persuasive message introducing a new company. The message was composed of either strong or weak arguments. Subsequently, source status was manipulated by attributing the message to a source in the numerical minority or majority. As predicted, thought confidence was higher when the source had majority rather than minority status. As a consequence, the majority endorsement increased persuasion when participants had positive thoughts (strong argument condition), but decreased persuasion when participants had negative thoughts (weak argument condition). Again, though, the effect of source status on thought confidence occurred only when source information followed information processing and when elaboration was high (Horcajo, Briñol, & Petty, 2014).

In addition to affecting the confidence people have in their thoughts about a message, source factors can affect the certainty with which people hold their attitudes following a message. For example, recent work indicates that attitude certainty is greater following persuasive messages from high rather than low credibility sources (e.g., Clarkson, Tormala, & Rucker, 2008), even when people resist those messages (Tormala & Petty, 2004a). For instance, if a

consumer receives a counterattitudinal message from an expert (as opposed to a non-expert), the consumer can infer that he or she now has access to the best information on the topic. Regardless of whether the consumer counterargues or accepts the message, having access to the most accurate or valid information on the topic tends to augment attitude certainty. Thus, even in the absence of attitude change as it is traditionally defined (in terms of valence and extremity), the credibility of the source of a message can shape meaningful attitudinal outcomes.

The numerical status of a source also affects attitude certainty. For example, although people typically resist persuasive messages from numerical minorities, research by Tormala, DeSensi, and Petty (2007) suggests that this resistance can mask underlying decreases in attitude certainty. The logic is that derogating (or writing off) a message source simply because that source is in the minority is perceived to be an illegitimate thing to do. Therefore, when people resist for that reason, their attitudes remain intact (e.g., they opposed some policy or product, received a message from a minority source, and then continue to oppose it), but they feel less certain about the attitude because they perceive that they have resisted attitude change for an illegitimate reason (see Rucker et al., 2014, for further discussion of legitimacy and certainty).

Message Effects on Certainty

As noted earlier, whether a message argues against or in favor of a given position is one of the most basic dimensions along which it can vary. Although there has not been much consideration of whether positive or negative messages generate greater certainty, there is some evidence that inducing people to think of their attitudes in terms of what or who they oppose, rather than support, leads to greater attitude certainty and more resistance to subsequent change (Bizer, Larsen, & Petty, 2011; Bizer & Petty, 2005). In addition, there is a considerable body of work examining the effects of learning that others oppose or support one's opinion. In particular, people feel more certain about their thoughts and attitudes when they receive high as opposed to low social consensus feedback – that is, when they learn that others agree rather than disagree with their thoughts and evaluations (e.g., Petrocelli, Tormala, & Rucker, 2007; Petty, Briñol, & Tormala, 2002; Visser & Mirabile, 2004). Interestingly, though, recent research suggests that this well-established effect of social consensus on certainty is moderated by one's need for assimilation versus uniqueness (Clarkson, Tormala, Rucker, & Dugan, 2013). Specifically, high consensus fosters greater certainty when people want to belong with or assimilate to others, but low consensus can foster greater certainty when people desire more uniqueness or differentiation.

Relatedly, people also tend to hold their attitudes with greater certainty when they perceive them to be based on consideration of both sides of an issue – that is, on both the pros and the cons (Rucker & Petty, 2004; Rucker, Petty, & Briñol, 2008). Even when the valence and extremity of their attitudes are no

different (e.g., because considering the cons did not uncover any negatives), merely perceiving that both sides of an issue have been weighed boosts the perception that one's information is more complete. At the same time, if considering both sides of an issue actually reveals conflicting information, such as a favorable product attribute but also a negative customer review, attitude certainty can be lowered as the accuracy of any individual piece of information becomes more questionable (e.g., Koriat, 2012; Smith, Fabrigar, MacDougall, & Wiesenhal, 2008). Thus, perceived consideration of both sides can foster greater certainty as long as that consideration does not yield too much contradictory evidence on a topic.

Argument quality also has been shown to influence attitude certainty. For example, Tormala and Petty (2002; 2004b) found that when people resisted persuasion, the stronger the persuasive message appeared to be, the more certain recipients felt of their original attitudes. In essence, resisting a compelling message is more impressive than resisting a weak or feeble message, leading people to feel more certain about the validity of their original attitudes when those attitudes withstand a strong rather than weak attack. In fact, this effect emerged even when participants across conditions resisted identical messages, but were led to believe that it was strong or weak.

Finally, the amount of information contained in a persuasive message can affect attitude certainty as well. In general, the more information people believe they have received from a message, the more certain they are of their post-message attitudes (Smith et al. 2008). This result has also been observed in research on omission neglect (e.g., Sanbonmatsu, Kardes, & Herr, 1992), which reveals that alerting people to potentially missing information from a message reduces the certainty with which they hold their attitudes. Reduced certainty, in turn, can foster a reluctance to act on one's attitude in the future (see Rucker et al., 2014).

Recipient Effects on Certainty

Just as characteristics of the source and message can affect psychological certainty, so too can characteristics of the target or recipient of the message. Unlike message factors, however, where the bulk of the research attention was on attitude certainty, with recipient factors the lion's share has been focused on understanding the confidence with which people hold their individual message-relevant thoughts.

One prominent feature of message recipients that has received attention is how people behave physically, or what they do with their bodies (see Briñol & Petty, 2008). To summarize, research shows that recipients' body postures, facial expressions, and physical movements can influence persuasion not only by serving as simple cues or affecting the direction and amount of thoughts, but also by affecting thought confidence. As with other sources of thought confidence, the confidence that emerges from behaviors or bodily states can moderate the effect of just about any thought that is currently available or on one's

mind (for a review on embodied validation, see Briñol, Petty, & Wagner, 2012). For instance, in a series of studies Briñol and Petty (2003) found that under high thinking conditions, head movements (e.g., nodding or shaking) affected the confidence people had in their thoughts, which mediated the effect of head movements on persuasion. Consequently, when people generated positive thoughts toward a proposal (e.g., while listening to strong arguments), vertical head movements (i.e., head nodding) led to more favorable attitudes than horizontal head movements (i.e., head shaking). However, when people listened to weak arguments and generated mostly negative thoughts toward the proposal, head nodding led to less favorable attitudes than head shaking. Subsequent research replicated these findings using body postures associated with confidence (e.g., pushing the chest out) versus doubt (e.g., hunching forward with a back curved; Briñol, Petty, & Wagner, 2009).

Research has revealed that emotions can play a similar role in persuasion by affecting thought confidence. For example, Briñol, Petty, and Barden (2007) found that when placed in a happy (versus sad) state following message processing, people felt more confident about their thoughts and relied on them more closely in forming their attitudes and behavioral intentions. Importantly, happiness can validate thoughts by increasing confidence in or liking of them. Thus, happiness increased persuasion when people had positive thoughts, but it decreased persuasion when people had negative thoughts. Replicating other self-validation studies, Briñol, Petty, and Barden (2007) also found support for the idea that emotion effects on thought confidence were restricted to high elaboration conditions (e.g., high need for cognition) and situations in which the emotion induction followed rather than preceded one's thinking. In contrast, under low elaboration conditions (e.g., low need for cognition), emotions had just a main effect on attitudes such that happy participants agreed with the message more than did sad participants. Subsequent research has replicated these findings with more unpleasant emotions that are also associated with confidence, such as anger and disgust (Petty & Briñol, 2015).

Bodily movements, behaviors, and emotions are not the only recipient variables relevant to persuasion. As introduced earlier, more cognitive feelings can also play an important role. For example, the subjective sense of ease with which new information can be perceived or generated has been shown to play an important role in persuasion. Indeed, ease effects – in particular, increased persuasion following the generation of two rather than ten positive arguments on a topic – can also be driven by thought confidence under high elaboration conditions (Tormala, Falces, Briñol, & Petty, 2007; Tormala, Petty, & Briñol, 2002). Moreover, other recipient variables have been studied with regard to self-validation, including self-affirmation, self-confidence, and power (see Briñol & Petty, 2009b, for a review). The confidence that emerges from the fit or matching of two related variables (e.g., Clark, et al., 2013; Huntsinger, 2013) and the confidence that sometimes is associated with responses to threat and aggression (Blankenship, Nesbit, & Murray, 2013; Briñol, Petty, & DeMarree, 2015) can also validate thoughts. As noted, in all of these cases, confidence in positive

thoughts increases persuasion while confidence in negative thoughts reduces persuasion.

In addition to the work just reviewed on thought confidence, there was some research exploring recipient factors that affect attitude certainty. As just one example, there is now considerable evidence that perceived thoughtfulness can influence the certainty with which people hold their attitudes. In general, the more people believe they have thought about or analyzed an object or issue, the more certain they are of their attitude toward it (Barden & Petty, 2008; Barden & Tormala, 2014; Wan, Rucker, Tormala, & Clarkson, 2010). For instance, Wan and colleagues (2010) presented participants with advertisements under conditions of regulatory depletion or non-depletion. Results indicated that when participants were depleted (versus non-depleted), they inferred that they had processed the ads more deeply, and this fostered greater attitude certainty. Interestingly, though, these effects can be moderated by other factors. For instance, Tormala, Clarkson, and Henderson (2011) found that attitude certainty increases when people perceive that they have taken their time and evaluated something slowly as long as the attitude object is unfamiliar. For familiar objects, people may have a different lay theory that reverses the weight given to slow versus fast evaluation.

Remaining Questions and Future Directions

As we have reviewed, attitude change research has a very rich history dating back many decades. Nonetheless, the field continues to evolve, particularly in our collective understanding of the varied and nuanced effects of classic persuasion variables and the mechanisms they trigger. Indeed, over the years, we have moved from the consideration of singular main effect predictions (e.g., that source credibility boosts persuasion) to more complex interaction predictions stemming from new insights around the cognitive processes driving persuasion (e.g., that source credibility can increase or decrease persuasion due to its effect on information processing) to still higher-order predictions for new outcomes based on the discovery of important metacognitive factors in this domain (e.g., showing that thought confidence and attitude certainty can be affected by people's perceptions of source credibility even when their attitude valence and extremity hold constant). Despite, or perhaps because of, this long history and constant evolution, there remain many open questions and opportunities for future research. In this final section of the chapter, we highlight what we see as some of the more interesting and promising next steps.

Before we do, we acknowledge that there have been other recent calls for future research. For example, Briñol and Petty (2012) described four areas of future research in attitudes and persuasion: (1) identifying more moderating conditions for effects and processes (e.g., moving from elaboration and timing to new contextual features), (2) studying new potential consequences of the different mechanisms of attitude change, (3) discovering new underlying

processes of persuasion, and (4) moving from single measures to multiple measures of attitude change. In the latter area, as noted earlier, the impact of persuasion treatments on attitudes first used deliberative measures, whereas now attitude change can also be assessed using automatic measures (Briñol, Petty, & McCaslin, 2009; Smith, De Houwer, & Nosek, 2013). Future research is likely to continue to explore this topic and take advantage of technological advances to understand attitudes. One area that is likely to see an exponential increase in interest concerns how attitudinal processes can be charted with new brain imaging techniques (e.g., Cunningham, Packer, Kesek, & Van Bavel, 2009).

In the remainder of the chapter, we focus on substantive areas and conceptual questions that we see as important to explore in the near future. Interestingly, these questions also key in on the source-message-recipient trichotomy outlined earlier. In particular, we submit that some of the more intriguing directions for new research relate to understanding source-message-recipient dynamics – that is, the interdependent and shifting nature of these variables. For instance, what happens to the source's attitude after delivering a persuasive message? Under what circumstances, and through what mechanisms, do message recipients become future message sources (e.g., advocates for a cause)? When and why do people deliberately act as both source and recipient and design messages to persuade themselves? What determines the kinds of messages that sources elect to deliver to others? We elaborate on these and other questions next.

Source–Recipient Dynamics

One important question that has received some but not nearly enough attention in past research relates to the interdependent relationships between message sources and recipients. For instance, how is the source of a message influenced by the apparent success or failure of that message in changing the recipient's attitude? Building on recent work investigating metacognitive factors in resistance and persuasion (Tormala, 2008), it stands to reason that when the source of a message can perceive the success or failure of his or her persuasion attempt, that perception could influence his or her attitude certainty, openness to future persuasion, and so on. Consistent with this logic, in a classic study in the communications domain, Infante (1976) found that after succeeding (versus failing) to persuade someone, people became more resistant to subsequent counterattitudinal proposals. More recently, Prislin and colleagues (2011) found that a source's initial success or failure in persuading others affected not only his or her certainty, but also his or her actual persuasive efficacy in subsequent persuasion attempts.

We see this area as ripe for further inquiry. As one possibility, just like one's own persuasion or resistance experience can lead one to reflect on one's attitude and adjust certainty upward or downward, we presume there are factors that foster certainty increases and decreases following both successful and unsuccessful attempts at persuading others. Perhaps if one can attribute one's success to the cogency of one's message, for instance, one gains certainty. If one

attributes one's success to situational factors (e.g., the recipient's good mood, or known status as a "yes man"), this effect might disappear or even induce doubt. Also interesting to explore is whether there are interpersonal effects of successful and unsuccessful persuasion attempts. Imagine that a source succeeds in persuading a recipient. Could this success be interpreted as reflecting that the source and recipient share core values and/or think in similar ways? If so, it seems reasonable to surmise that the source might like the recipient more after a successful (versus failed) persuasion attempt. In any case, the downstream consequences of persuasion attempts for the source, and perhaps his or her relationship with the recipient, are interesting and understudied topics in persuasion research.

Viewing source–recipient relations in a different light, persuasion can also be construed in terms of the dynamic relationship between two or more people. As noted, the bulk of the literature on persuasion to date has focused on how either an individual source or an individual recipient generates or responds to a persuasive message. One new and potentially important direction is to investigate how source and recipient variables interact with each other. Consider the work on power. On the face of it, given that high power represents a position of dominance over low power, it seems reasonable to expect more persuasion when the source has power over the recipient (e.g., compliance; Kelman, 1958) and less persuasion when the recipient has power over the source. While this certainly is possible (see Lammers, Dubois, Rucker, & Galinsky, 2013), it is also possible that greater persuasion arises when there is a match between the power of the source and recipient.

Consistent with this latter notion, Dubois, Rucker, and Galinsky (2014) recently found that high power sources are sometimes more persuasive when delivering their messages to high power recipients, whereas low power sources tend to be more persuasive when delivering their messages to low power recipients. In other words, Dubois and colleagues observed a matching effect with power and persuasion. Importantly, matching, like any other variable, can operate through multiple processes (e.g., affecting processing, validating thoughts), thus increasing or decreasing persuasion depending on the circumstances (Briñol, Tormala, & Petty, 2013).

Recipient (Source) Variables as Source (Recipient) Variables

Also germane to understanding that sources and recipients can vary along the same dimensions, there are a multitude of variables that have been examined primarily with respect to the recipient but also apply to the source, and that have been examined primarily with respect to the source but also apply to the recipient. Future research would benefit from expanding current conceptualizations of recipient variables to source variables and vice versa.

As one example, individual differences in need for cognition (the tendency to enjoy and engage in effortful thought; Cacioppo & Petty, 1982) have been studied almost exclusively as a recipient variable. Some recipients think more

than others, and this difference influences the way they process persuasive messages (see Cacioppo, Petty, Feinstein, & Jarvis, 1996). However, sources presumably vary in need for cognition as well. How does this variation affect their persuasiveness? Briñol et al. (2005) found that although high (versus low) need for cognition individuals were able to generate more convincing arguments in a group setting, they were less efficient at reaching group consensus as the size of the group increased. The authors observed that high need for cognition individuals tended to easily persuade those low in need for cognition because of their greater arsenal of arguments and enhanced resistance to counterarguments (Shestowsky, Wegener, & Fabrigar, 1998), but they had no persuasive advantage against others who shared their high need for cognition, unless they received training on group dynamics (see Petty, Briñol, Loersch, & McCaslin, 2009, for a review).

On a related note, Kupor, Tormala, Norton, and Rucker (2014) recently suggested that although thoughtfulness traditionally has been studied as a recipient variable (e.g., more thoughtfulness creates stronger attitudes; see Barden & Petty, 2008), it can also be explicitly high or low in a message source (e.g., “I have given this a lot of thought and. . .” or “My quick gut reaction is. . .”). In a series of studies, Kupor et al. (2014) found that people liked others more, and were more influenced by their decisions, when those others calibrated their level of thoughtfulness to the demands of the situation. When making a difficult decision, for instance, more source thoughtfulness fostered greater influence and liking. When making an easy decision, however, less thoughtfulness fostered greater influence and liking.

Recent work has taken a similar approach to attitude certainty. Again, although certainty has been studied almost exclusively with respect to message recipients (e.g., high certainty increases resistance to persuasion; Tormala & Rucker, 2007), it can vary on the source side as well. How does a source’s expressed certainty or uncertainty affect persuasion? In a recent series of studies, Karmarkar and Tormala (2010) presented participants with restaurant reviews that explicitly incorporated some statement of certainty or uncertainty. For instance, some reviews gave the restaurant a “confident 4 out of 5 stars,” whereas others gave the restaurant a “tentative 4 out of 5 stars.” Interestingly, results indicated that although non-expert sources could increase their persuasiveness by expressing certainty rather than uncertainty, the opposite was true for expert sources. That is, experts gained persuasiveness by expressing less rather than more attitude certainty. The authors found that when expert (non-expert) sources expressed less (more) certainty, that struck recipients as surprising or unexpected, causing them to tune in and process the message more deeply, which gave it more impact.

Going forward, there is a great opportunity for researchers to continue this line of inquiry. Theoretically, any variable that traditionally has been viewed as a recipient variable could also be studied as a source variable. This might include other forms of attitude strength such as attitude importance, accessibility, or ambivalence; mood states; personality variables; and so on. Similarly,

any variable that traditionally has been studied as a source variable could be subject to research as a recipient variable, including expertise, trustworthiness, attractiveness, likeability, and the like. Illustration of this latter possibility come from work on epistemic authority in which source credibility is examined as a recipient rather than a source variable (Kruglanski et al., 2005), research on individual differences in self-monitoring, which have been studied as both a recipient and a source variable (Shavitt, Lowrey, & Han, 1992), and work on numerical status, with minority status being relevant as both a source and recipient variable (Morrison, 2011). We see this as an interesting and fruitful direction for future work.

Turning Recipients into Sources: Creating Advocacy

Another important yet understudied topic in the persuasion literature pertains to the generation of persuasive advocacy. Stated differently, what factors determine whether the recipient of a persuasive message will become a future *source* of a related message? What transforms an attitude holder into an attitude advocate? This topic has remained relatively underdeveloped over the years despite its obvious importance, particularly in consumer psychology, where word of mouth and sharing behavior have emerged as central topics (see Berger, 2014). However, there has been some classic and contemporary work that is of at least indirect relevance, including research on attitude bolstering (Briñol, Rucker, Tormala, & Petty, 2004; Lydon, Zanna, & Ross, 1988; Xu and Wyer, 2012), research on supportive resistance strategies (McGuire, 1964; Visser, Krosnick, & Simmons, 2003), studies on the persuasive effect of the transmitter (Echterhoff, Higgins, Kopietz, & Groll, 2008), work on the “saying is believing” effect (Hausmann, Levine, & Higgins, 2008), and research on proselytizing (Festinger, Riecken, & Schachter, 1956; Gal & Rucker, 2010).

In one recent study exploring the determinants of advocacy, Akhtar, Paunesku, and Tormala (2013) examined the potential effects of message strength. The authors presented prospective voters with a strong or weak persuasive message endorsing their preferred candidate in the 2012 presidential election (e.g., pro-Obama participants received a message in favor of reelecting Barack Obama), and subsequently asked about their advocacy intentions – for instance, their willingness to help the Obama campaign and to try to persuade others to share their view on Obama. Ironically, results indicated that recipients expressed greater advocacy intentions, and actually wrote more in an advocacy message, when they first received weak rather than strong arguments supporting their existing positions. In essence, weak proattitudinal messages induced recipients to believe that they had something important and unique to contribute to the cause, which turned them into advocates, or more willing and energized sources of future messages in the same direction.

Further research that delineates the key contributors to advocacy would help illuminate the factors that transform message recipients (e.g., supporters of a

cause) into message sources (e.g., advocates for that cause). In so doing, future studies could help deepen our insight into the numerous important outcomes stemming from persuasive messages that reach beyond attitudes themselves. Indeed, in the Akhtar, Paunesku, and Tormala (2013) research, weak arguments were not more *persuasive* than strong ones. In fact, because participants already held relatively firm attitudes toward Barack Obama, message strength had no effect on attitudes in those studies. Importantly, though, it had a clear and consistent effect on attitudinal *advocacy*. Thus, by better understanding the subtle and previously hidden effects of persuasion variables, we can better predict and apply those effects in the service of important attitude-relevant objectives beyond traditionally identified metrics of persuasion (e.g., attitude measures).

Also important, once an individual decides to advocate for a cause, or attempt to persuade someone else, what determines the strategy he or she employs? As reviewed in this chapter, we know much about the effects of different persuasion strategies on the effectiveness of a given appeal, but we know very little about the determinants of the strategies that sources elect to implement. We assume this decision is guided at least in part by people's lay theories and persuasion knowledge (Briñol, Rucker, & Petty, 2015; Friestad & Wright, 1994, 1995; Kirmani & Campbell, 2004), but what are those theories and how do they vary by person and context? This question is wide open for future exploration and could be of tremendous theoretical and practical import.

Partly related to our previous discussion, how do sources' persuasion strategies affect their own attitudes and self-perceptions? An interesting line of research by Kipnis and colleagues suggests that using different social influence strategies can have different effects. For example, when people rely on their authority or power to influence others, they tend to feel worse about themselves, about their persuasive attempts, and about the recipients than they do when they convince others using arguments or other impression formation strategies that do not exploit their power or authority (Kipnis, Schmidt, & Wilkinson, 1980; O'Neal, Kipnis, & Craig, 1994; Rind & Kipnis, 2002). In general, it could be that some persuasion strategies are perceived as more legitimate than others, and that sources feel more certain about their own attitudes, and about themselves, when they use seemingly legitimate means (for a related discussion, see Tormala DeSensi, & Petty, 2007). Future research could enhance our understanding of both source and recipient persuasion dynamics by exploring these issues.

Turning Sources into Recipients: Self-Persuasion

Finally, one of the classic and most indispensable topics in attitude change and persuasion research is that of self-persuasion: understanding the situations, motives, and mechanisms that lead people to persuade themselves to adopt a new attitude. At its core, self-persuasion is about inducing people to serve as the

source of a message aimed at themselves (see Maio & Thomas, 2007). In this respect, self-persuasion is about turning message recipients into their own message sources. One classic study of self-persuasion explored the effects of role playing on attitude change (Janis & King, 1954). In this initial work, participants were induced to act out a specified role (e.g., convince a friend to stop smoking or listen to an appeal to stop smoking). Famously, participants were more persuaded in the direction of the message when they generated it themselves compared to when they passively observed its delivery or listened to the information. One account for this effect is that self-generated messages are more persuasive because creating them leads people to engage in biased scanning of the arguments that they find to be especially compelling (e.g., Greenwald & Albert, 1968).

More recently, self-persuasion research has adopted a metacognitive perspective. For instance, Briñol, McCaslin, and Petty (2012) found that people put more effort into generating persuasive messages when they had doubts rather than confidence in a point of view (see also Gal & Rucker, 2010; cf. Akhtar, Paunesku, & Tormala, 2013; Matthes, Morrison, & Schemer, 2010; Rios, DeMarree, & Statzer, 2014). Greater effort in generating the message led to more self-persuasion. Also germane, recent research on the mere thought effect – the phenomenon whereby merely thinking about an object or issue can foster attitude polarization because people bring attitude-consistent thoughts to mind (e.g., Tesser, 1978) – suggests that self-persuasion effects can be amplified under conditions in which people feel greater confidence. For example, Clarkson, Tormala, and Leone (2011) varied the amount of time people had to think about an issue on which they initially held moderate attitudes. When participants had more time to think, they held their (attitude-consistent) thoughts with more confidence, and thus showed greater self-generated attitude change. Interestingly, though, too much time to think undid this effect, as people began to doubt their own thoughts and depolarize their attitudes (see also Clarkson, Valente, Leone, & Tormala, 2013).

Particularly in the domain of consumer behavior, we believe this classic topic of self-persuasion is ready for a rebirth. Indeed, it is now widely reported that consumers are growing increasingly skeptical of conventional marketing and advertising messages, relying more and more on word of mouth and other consumers for their information and opinions (see Dubois, Rucker, & Tormala, 2011). Accordingly, marketers have made more and more use of social networks, customer reviews, and alternative information sources to spread their messages. A natural next step, we postulate, is to create the conditions under which consumers will be persuaded by themselves or their own thinking. Consider an example in which an individual consumer selects an item (e.g., a pair of shoes) from an online retailer and then transfers to a checkout or purchase page. Based on the findings of Clarkson, Tormala, and Leone (2011), there might be some advantage to getting this consumer to wait briefly, which could allow him or her to generate more attitude-consistent thoughts (e.g., “those shoes really will be great for going out at night”), which in turn

boosts his or her likelihood of completing the purchase and ordering the shoes (see Clarkson, Tormala, & Duhachek, 2010).

Relatedly, it is well known that merely completing a customer satisfaction survey can enhance customer satisfaction (Dholakia & Morwitz, 2002). Perhaps customer satisfaction surveys represent opportunities for customer influence via self-persuasion. If, for example, customers were asked to articulate the reasons for their purchase or explain the ways in which the product has been useful to them, a self-persuasion process may be engaged in which customers selectively focus on favorable arguments supporting their purchase and thereby convince themselves that it was a good one. Indeed, inducing people to list positive or negative thoughts on a topic can foster more positive and negative attitudes, respectively (Killeya & Johnson, 1998; Xu & Wyer, 2012). Exploring these kinds of effects in customer survey scenarios could be a useful next step in this domain.

More broadly, we see self-persuasion as an extremely important and still underexplored area in the attitudes literature. Although it had a prominent position in early attitudes research, it has been relatively neglected by recent scholars. There remain many open questions. For instance, when and why do people seek to persuade themselves? Is it when they have discrepancies between their actual and desired attitudes (DeMarree, Wheeler, Briñol, & Petty, 2014)? What strategies do people use? Are some strategies more effective than others? What contextual factors turn these strategies on and off? There are, of course, many more questions that could be raised. For now, we simply highlight self-persuasion as an area of future growth, with many open questions of both a basic and an applied nature.

Coda

Attitude change and persuasion are at the heart of consumer behavior. Research in this area dates back to the middle of the twentieth century. Since that time, attitudes, attitude change, and persuasion have been among the most studied topics in social and consumer psychology, and perhaps in psychology more generally. In this chapter, we sought to provide a broad overview of the persuasion literature, covering classic views of important variables and their effects, discussing modern metacognitive reconceptualizations, and highlighting unanswered or just partially answered questions deserving of new and deeper consideration. Over the years, researchers have made great strides in advancing our basic understanding of persuasion variables and when and how they exert their influence. Nevertheless, the field is constantly evolving and there remain many exciting paths forward. Attitudes are and will continue to be one of the most indispensable constructs in consumer research, and we hope the current chapter can serve as at least a partial guide, and strong encouragement, to pursue some of the important and exciting next steps.

References

- Abelson, R. P. (1995). Attitude extremity. In R. E. Petty & J. A. Krosnick (eds.), *Attitude Strength: Antecedents and Consequences* (pp. 25–42). Mahwah, NJ: Erlbaum.
- Akhtar, O., Paunesku, D., & Tormala, Z. L. (2013). Weak > strong: The ironic effect of argument strength on supportive advocacy. *Personality and Social Psychology Bulletin*, 39, 1214–1226.
- Barden, J., & Petty, R. E. (2008). The mere perception of elaboration creates attitude certainty: Exploring the thoughtfulness heuristic. *Journal of Personality and Social Psychology*, 95, 489–509.
- Barden, J., & Tormala, Z. L. (2014). Elaboration and attitude strength: The new meta-cognitive perspective. *Social and Personality Psychology Compass*, 8, 17–29.
- Bassili, J. N. (1996) Meta-Judgmental versus operative indices of psychological properties: The case of measures of attitude strength. *Journal of Personality and Social Psychology*, 71, 637–653.
- Berger, J. (2014). Word-of-mouth and interpersonal communication: An organizing framework and directions for future research. *Journal of Consumer Psychology*, 24, 586–607.
- Bizer, G. Y., Larsen, J. T., & Petty, R. E. (2011). Exploring the valence-framing effect: Negative framing enhances attitude strength. *Political Psychology*, 32, 59–80.
- Bizer, G. Y., & Petty, R. E. (2005). How we conceptualize our attitudes matters: The effects of valence framing on the resistance of political attitudes. *Political Psychology*, 26, 553–568.
- Blankenship, K. L., Nesbit, S. M., & Murray, R. A. (2013). Driving anger and meta-cognition: The role of thought confidence on anger and aggressive driving intentions. *Aggressive Behavior*, 39, 323–334.
- Bohner, G., Ruder, M., & Erb, H.-P. (2002). When expertise backfires: Contrast and assimilation effects in persuasion. *British Journal of Social Psychology*, 41, 495–519.
- Boninger, D. S., Krosnick, J. A., Berent, M. K., & Fabrigar, L. R. (1995). The causes and consequences of attitude importance. In R. E. Petty & J. A. Krosnick (eds.), *Attitude Strength: Antecedents and Consequences* (pp. 159–190). Mahwah, NJ: Erlbaum.
- Bornstein, R. F. (1989). Exposure and affect: Overview and meta-analysis of research, 1968–1987. *Psychological Bulletin*, 106, 265–289.
- Briñol, P., Becerra, A., Díaz, D., Horcajo, J., Valle, C., & Gallardo, I. (2005). The impact of need for cognition on interpersonal influence. *Psicothema*, 17, 666–671.
- Briñol, P. & DeMarree, K. G., eds. (2012) *Social Metacognition*. New York: Psychology Press.
- Briñol, P., McCaslin, M. J., & Petty, R. E. (2012). Self-generated persuasion: Effects of the target and direction of arguments. *Journal of Personality and Social Psychology*, 102, 925–940.
- Briñol, P. & Petty, R. E. (2003) Overt head movements and persuasion: A self-validation analysis. *Journal of Personality and Social Psychology*, 84, 1123–1139.
- Briñol, P. & Petty, R. E. (2005). Individual differences in persuasion. In D. Albarracín, B. T. Johnson, & M. P. Zanna (eds.), *The Handbook of Attitudes and Attitude Change* (pp. 575–616). Hillsdale, NJ: Erlbaum.

- Briñol, P. & Petty, R. E (2008). Embodied persuasion: Fundamental processes by which bodily responses can impact attitudes. In G. R. Semin & E. R. Smith (eds.), *Embodiment Grounding: Social, Cognitive, Affective, and Neuroscientific Approaches* (pp. 184–207). Cambridge: Cambridge University Press.
- Briñol, P. & Petty, R. E (2009a). Source factors in persuasion: A self-validation approach. *European Review of Social Psychology*, 20, 49–96.
- Briñol, P. & Petty, R. E (2009b). Persuasion: Insights from the self-validation hypothesis. In M. P. Zanna (ed.), *Advances in Experimental Social Psychology*, 41, (pp. 69–118). New York: Academic Press.
- Briñol, P. & Petty, R. E (2012). The history of attitudes and persuasion research. In A. Kruglanski & W. Stroebe (eds.), *Handbook of the History Of Social Psychology* (pp. 285–320). New York: Psychology Press.
- Briñol, P., Petty, R. E., & Barden, J. (2007). Happiness versus sadness as determinants of thought confidence in persuasion: A self-validation analysis. *Journal of Personality and Social Psychology*, 93, 711–727.
- Briñol, P., Petty, R. E., & DeMarree, K. (2015). Being threatened and being a threat can increase reliance on thoughts: A self-validation approach. In P. J. Carroll, R. M. Arkin, & A. Wichman (eds.), *Handbook on Personal Security* (pp. 37–54). New York: Psychology Press.
- Briñol, P., Petty, R. E., & McCaslin, M. (2009). Changing attitudes on implicit versus explicit measures: What is the difference? In R. E. Petty, R. H. Fazio, & P. Briñol (eds.), *Attitudes: Insights from the New Implicit Measures* (pp. 285–326). New York: Psychology Press.
- Briñol, P., Petty, R. E., & Tormala, Z. L. (2004). The self-validation of cognitive responses to advertisements. *Journal of Consumer Research*, 30, 559–573.
- Briñol, P., Petty, R. E. & Wagner, B. C. (2009) Body postures effects on self-evaluation: A self-validation approach. *European Journal of Social Psychology* 39, 1053–64.
- Briñol, P., Petty, R. E. & Wagner, B. C. (2012). Embodied validation: Our body can change and also validate our thoughts. In P. Briñol & K. G. DeMarree (eds.), *Social Metacognition* (pp. 219–240). New York: Psychology Press.
- Briñol, P., Petty, R. E., & Wheeler, S. C. (2006). Discrepancies between explicit and implicit self-concepts: Consequences for information processing. *Journal of Personality and Social Psychology*, 91, 154–170.
- Briñol, P., Rucker, D. D., & Petty, R. E. (2015). Naïve theories about persuasion: Implication for information processing and consumer attitude change. *International Journal of Advertising*, 34, 85–106.
- Briñol, P., Rucker, D., Tormala, Z. L., & Petty, R. E. (2004). Individual differences in resistance to persuasion: The role of beliefs and meta-beliefs. In E. S. Knowles & J. A. Linn (eds.), *Resistance and Persuasion* (pp. 83–104). Mahwah, NJ: Erlbaum.
- Briñol, P., Tormala, Z. L., & Petty, R. E. (2013). Ease and persuasion: Multiple processes, meanings, and effects. In C. Unkelbach & R. Greifeneder (eds.), *The Experience of Thinking: How the Fluency of Mental Processes Influences Cognition and Behavior* (pp. 101–118). London: Psychology Press.
- Byrne, D. (1971). *The Attraction Paradigm*. New York: Academic Press
- Cacioppo, J. T., & Petty, R. E. (1979). The effects of message repetition and position on cognitive responses, recall, and persuasion. *Journal of Personality and Social Psychology*, 37, 97–109.

- Cacioppo, J. T., & Petty, R. E. (1982). The need for cognition. *Journal of Personality and Social Psychology*, 42, 116–131.
- Cacioppo, J. T., & Petty, R. E. (1989). Effects of message repetition on argument processing, recall, and persuasion. *Basic and Applied Social Psychology*, 10, 3–12.
- Cacioppo, J. T., Petty, R. E., Feinstein, J. A., & Jarvis, W. B. G. (1996). Dispositional differences in cognitive motivation: The life and times of individuals varying in need for cognition. *Psychological Bulletin*, 119, 197–253.
- Cacioppo, J. T., Priester, J. R., & Berntson, G. G. (1993). Rudimentary determinants of attitudes II: Arm flexion and extension have differential effects on attitudes. *Journal of Personality and Social Psychology*, 65, 5–17.
- Calder, B. J., Insko, C., & Yandell, B. (1974). The relation of cognitive and memorial processes to persuasion in a simulated jury trial. *Journal of Applied Social Psychology*, 4, 62–93.
- Chaiken, S. (1980). Heuristic versus systematic information processing in the use of source versus message quest in persuasion. *Journal of Personality and Social Psychology*, 39, 752–766.
- Chaiken, S., & Ledgerwood, A. (2012). A theory of heuristic and systematic information processing. In P. A. M. van Lange, A. W. Kruglanski, & E. T. Higgins (eds.), *Handbook of Theories of Social Psychology* (pp. 246–266). Thousand Oaks, CA: Sage.
- Chaiken, S. L., Liberman, A., & Eagly, A. H. (1989). Heuristic and systematic processing within and beyond the persuasion context. In J. S. Uleman y J. A. Bargh (eds.), *Unintended Thought* (pp. 212–252). New York: Guilford Press.
- Chaiken, S., & Maheswaran, D. (1994). Heuristic processing can bias systematic processing: Effects of source credibility, argument ambiguity, and task importance on attitude judgment. *Journal of Personality and Social Psychology*, 66, 460–473.
- Clark, J. K., & Wegener, D. T. (2013). Message position, information processing, and persuasion: The Discrepancy Motives Model. In P. Devine & A. Plant (eds.), *Advances in Experimental Social Psychology* (vol. 47, pp. 189–232). San Diego, CA: Academic Press.
- Clark, J. K., Wegener, D. T., Sawicki, V., Petty, R. E., & Briñol, P. (2013). Evaluating the message or the messenger? Implications for self-validation in persuasion. *Personality and Social Psychology Bulletin*, 39, 1571–1584.
- Clarkson, J. J., Tormala, Z. L., & Duhachek, A. L. (2010). Seeking optimality in the consumer waiting experience: The good and bad of waiting time. Paper presented at the annual meeting of the Society for Consumer Psychology, St. Petersburg, FL.
- Clarkson, J. J., Tormala, Z. L., & Leone, C. (2011). A self-validation perspective on the mere thought effect. *Journal of Experimental Social Psychology*, 47, 449–454.
- Clarkson, J. J., Tormala, Z. L., & Rucker, D. D. (2008). A new look at the consequences of attitude certainty: The amplification hypothesis. *Journal of Personality and Social Psychology*, 95, 810–825.
- Clarkson, J. J., Tormala, Z. L., Rucker, D. D., & Dugan, R. G. (2013). The malleable influence of social consensus on attitude certainty. *Journal of Experimental Social Psychology*, 49, 1019–1022.
- Clarkson, J. J., Valente, M. J., Leone, C., & Tormala, Z. L. (2013). Motivated reflection on attitude-inconsistent information: An exploration of the role of fear of

- invalidity in self-persuasion. *Personality and Social Psychology Bulletin*, 29, 1559–1570.
- Clore, G. L., & Parrott, W. G. (1994). Cognitive feelings and metacognitive judgments. *European Journal of Social Psychology (Special Issue)*, 24, 101–115.
- Cohen, J. B., & Reed, A. (2006). A multiple pathway anchoring and adjustment (MPAA) model of attitude generation and reinforcement. *Journal of Consumer Research*, 33, 1–15.
- Crowley, A. E., & Hoyer, W. D. (1994). An integrative framework for understanding two-sided persuasion. *Journal of Consumer Research*, 20, 561–574.
- Cunningham, W. A., Packer, D. J., Kesek, A., & Van Bavel, J. J. (2009). Implicit measurement of attitudes: A physiological approach. In R. E. Petty, R. H. Fazio, & P. Briñol (eds.), *Attitudes: Insights from the New Implicit Measures* (pp. 485–512). New York: Psychology Press.
- Davidson, A. R., Yantis, S., Norwood, M., & Montano, D. E. (1985). Amount of information about the attitude object and attitude–behavior consistency. *Journal of Personality and Social Psychology*, 49, 1184–1198.
- DeMarree, K. G., Wheeler, C. S., Briñol, P., & Petty, R. E. (2014). Wanting other attitudes: Actual–desired attitude discrepancies predict feelings of ambivalence and ambivalence consequences. *Journal of Experimental Social Psychology*, 53, 5–18.
- Dholakia, U. M., & Morwitz, V. G. (2002). The scope and persistence of mere measurement effects: Evidence from a field study of customer satisfaction measurement. *Journal of Consumer Research*, 29, 159–167.
- Dubois, D., Rucker, D. D., & Galinsky, A. D. (2014). The power matching effect: Psychological attunement between communicator and audience power enhances persuasion. Unpublished manuscript. INSEAD.
- Dubois, D., Rucker, D. D., & Tormala, Z. L. (2011). From rumors to facts, and facts to rumors: The role of certainty decay in consumer communications. *Journal of Marketing Research*, 48, 1020–1032.
- Eagly, A. H., & Chaiken, S. (1993). *The Psychology of Attitudes*. Fort Worth, TX: Harcourt, Brace, Jovanovich.
- Echterhoff, G., Higgins, T. E., Kopietz, R., & Groll, S. (2008). How communication goals determine when audience tuning biases memory. *Journal of Experimental Psychology: General*, 137, 3–21.
- Edwards, K. (1990). The interplay of affect and cognition in attitude formation and change. *Journal of Personality and Social Psychology*, 59, 202–216.
- Ein-Gar, D., Shiv, B., & Tormala, Z. L. (2012). When blemishing leads to blossoming: The positive effect of negative information. *Journal of Consumer Research*, 38, 846–859.
- Etgar, M., & Goodwin, S. A. (1982). One-Sided versus two-sided comparative message appeals for new brand introduction. *Journal of Consumer Research*, 8, 460–465.
- Fabrigar, L. R., & Petty, R. E. (1999). The role of the affective and cognitive bases of attitudes in susceptibility to affectively and cognitively based persuasion. *Personality and Social Psychology Bulletin*, 25, 363–381.
- Fazio, R. H. (1990). Multiple processes by which attitudes guide behaviour: The MODE model as an integrative framework. In M. P. Zanna (ed.), *Advances in Experimental Social Psychology* (vol. 23, pp. 75–109). San Diego, CA: Academic Press.

- Fazio, R. H. (1995). Attitudes as object-evaluation associations: Determinants, consequences, and correlates of attitude accessibility. In R. E. Petty & J. A. Krosnick (eds.), *Attitude Strength: Antecedents and Consequences* (pp. 247–282). Hillsdale, NJ: Erlbaum.
- Feldman, J. M., & Lynch, J. (1988). Self-generated validity and other effects of measurement on belief, attitude, intention, and behavior. *Journal of Applied Psychology*, 73, 421–435.
- Festinger, L., Riecken, H. W., & Schachter, S. (1956). *When Prophecy Fails*. Minneapolis, MN: University of Minnesota Press.
- Festinger, L., & Thibaut, J. (1951). Interpersonal communications in small groups. *Journal of Abnormal and Social Psychology*, 46, 92–100.
- Fishbein, M., & Ajzen, I. (1975). *Belief, Attitude, Intention, and Behavior*. Reading, MA: Addison-Wesley.
- Fleming, M. A., & Petty, R. E. (2000). Identity and persuasion: An elaboration likelihood approach. In D. J. Terry & M. A. Hogg (eds.), *Attitudes, Behavior, and Social Context: The Role of Norms and Group Membership* (pp. 171–199). Mahwah, NJ: Lawrence Erlbaum.
- French, J. R. P., Jr., & Raven, B. H. (1959). The bases of social power. In D. Cartwright (ed.), *Studies in Social Power* (pp. 150–167). Ann Arbor, MI: Institute for Social Research.
- Friestad, M., & Wright, P. (1994). The Persuasion Knowledge Model: How people cope with persuasion attempts. *Journal of Consumer Research*, 21, 1–31.
- Friestad, M., & Wright, P. (1995). Persuasion knowledge: Lay people's and researchers' beliefs about the psychology of persuasion. *Journal of Consumer Research*, 27, 123–156.
- Gal, D., & Rucker, D. D. (2010). When in doubt, shout! Paradoxical influences of doubt on proselytizing. *Psychological Science*, 20, 1–7.
- Gawronski, B., & Payne, B. K. (2010) (eds.). *Handbook of Implicit Social Cognition: Measurement, Theory, and Applications*. New York: Guilford Press.
- Gorn, G., & Goldberg, M. (1980). Children's responses to repetitive TV commercials. *Journal of Consumer Research*, 6, 421–425.
- Greenwald, A. G., & Albert, R. D. (1968). Acceptance and recall of improvised arguments. *Journal of Personality and Social Psychology*, 8, 31–34.
- Gross, S. R., Holtz, R., & Miller, N. (1995). Attitude certainty. In R. E. Petty & J. A. Krosnick (eds.), *Attitude Strength: Antecedents and Consequences* (pp. 215–245). Hillsdale, NJ: Erlbaum.
- Grush, J. E. (1976). Attitude formation and mere exposure phenomena: A non-artificial explanation of empirical findings. *Journal of Personality and Social Psychology*, 33, 281–290.
- Hausmann, L. R. M., Levine, J. M., & Higgins, E. T. (2008). Communication and group perception: Extending the “saying is believing” effect. *Group Processes and Intergroup Relations*, 11, 539–554.
- Heesacker, M. H., Petty, R. E., & Cacioppo, J. T. (1983). Field dependence and attitude change: Source credibility can alter persuasion by affecting message-relevant thinking. *Journal of Personality*, 51, 653–666.
- Horcajo, J., Briñol, P., & Petty, R. E. (2014). Multiple roles for majority versus minority source status on persuasion when source status follows the message. *Social Influence*, 9, 37–51.

- Horcajo, J., Petty, R. E., & Briñol, P. (2010). The effects of majority versus minority source status on persuasion: A self-validation analysis. *Journal of Personality and Social Psychology*, 99, 498–512.
- Hovland, C. I., Lumsdaine, A. A., & Sheffield, F. D. (1949). *Experiments on Mass Communication*. Princeton, NJ: Princeton University Press.
- Huntsinger, J. R. (2013). Incidental experiences of affective coherence and incoherence influence persuasion. *Personality and Social Psychology Bulletin*, 39, 792–802.
- Infante, D. A. (1976). Persuasion as a function of the receiver's prior success or failure as a message source. *Communication Quarterly*, 24, 21–26.
- Janis, I. L., & King, B. T. (1954). The influence of role-playing on opinion change. *Journal of Abnormal and Social Psychology*, 49, 211–218.
- Josephs, R. A., Giesler, R. B., & Silvera, D. H. (1994). Judgment by quantity. *Journal of Experimental Psychology: General*, 123, 21–32.
- Karmarkar, U.R., & Tormala, Z.L. (2010). Believe me, I have no idea what I'm talking about: The effects of source certainty on consumer involvement and persuasion. *Journal of Consumer Research*, 46, 1033–1049.
- Kaufman, D. Q., Stasson, M. F., & Hart, J. W. (1999). Are the tabloids always wrong or it that just what we think? Need for cognition and perceptions of articles in print media. *Journal of Applied Social Psychology*, 29, 1984–1997.
- Kelman, H. C. (1958). Compliance, identification and internalization: Three processes of attitude change. *Journal of Conflict Resolution*, 2, 51–60.
- Kelman, H. C., & Hovland, C. I. (1953). "Reinstatement" of the communicator in delayed measurement of opinion change. *Journal of Abnormal and Social Psychology*, 48, 327–335.
- Killeya, L. A. & Johnson, B. T. (1998). Experimental induction of biased systematic processing: The directed-thought technique. *Personality and Social Psychology Bulletin*, 24, 17–33.
- Kipnis, D., Schmidt, S. M., & Wilkinson, I. (1980). Intraorganizational influence tactics: Explorations in getting one's way. *Journal of Applied Psychology*, 65, 440–452.
- Kirmani, A., & Campbell, M. C. (2004). Goal seeker and persuasion sentry: How consumer targets respond to interpersonal marketing persuasion. *Journal of Consumer Research*, 31, 573–582.
- Koriat, A. (2012). The self-consistency model of subjective confidence. *Psychological Review*, 119, 80–113.
- Krosnick, J. A., & Petty, R. E. (1995). Attitude strength: An overview. In R. E. Petty & J. A. Krosnick (eds.), *Attitude Strength: Antecedents and Consequences* (pp. 1–24). Mahwah, NJ: Erlbaum.
- Kruglanski, A. W., Raviv, A., Bar-Tal, D., Raviv, A., Sharvit, K., Ellis, S., Bar, R., Pierro, A., & Mannetti, L. (2005). Says who? Epistemic authority effects in social judgment. In M. P. Zanna (ed.), *Advances in Experimental Social Psychology* (vol. 37, pp. 346–392). San Diego, CA: Academic Press.
- Kupor, D., Tormala, Z. L., Norton, M. I., & Rucker, D. D. (2014). Thought calibration: How thinking just the right amount increases one's influence and appeal. *Social Psychological and Personality Science*, 5, 263–270.
- Labroo, A. A., Dhar, R., & Schwarz, N. (2008). Of frog wines and frowning watches: Semantic priming, perceptual fluency, and brand evaluation. *Journal of Consumer Research*, 34, 819–831.

- Lammers, J., Dubois, D., Rucker, D. D., & Galinsky, A. D. (2013). Power gets the job: Priming power improves interview outcomes. *Journal of Experimental Social Psychology, 49*, 776–779.
- Lee, A. Y., & Labroo, A. A. (2004). The effect of conceptual and perceptual fluency on brand evaluation. *Journal of Marketing Research, 41*, 151–165.
- Lord, C., Ross, L., & Lepper, M. (1979). Biased assimilation and attitude polarization: The effects of prior theories on subsequently considered evidence. *Journal of Personality and Social Psychology, 37*, 2098–2109.
- Lydon, J., Zanna, M. P., & Ross, M. (1988). Bolstering attitudes by autobiographical recall: Attitude persistence and selective memory. *Personality and Social Psychology Bulletin, 14*, 78–86.
- Lynch, J. G. Jr. (2006). Accessibility-diagnosticsity and the multiple pathway anchoring and adjustment model. *Journal of Consumer Research, 33*, 25–27.
- Maio, G. R., & Thomas, G. (2007). The epistemic–teleologic model of deliberate self-persuasion. *Personality and Social Psychology Review, 11*, 46–67.
- Martin, L. L., Ward, D. W., Achee, J. W., & Wyer, R. S. (1993). Mood as input: People have to interpret the motivational implications of their moods. *Journal of Personality and Social Psychology, 64*, 317–326.
- Mayer, N. D., & Tormala, Z. L. (2010). “Think” versus “feel” framing effects in persuasion. *Personality and Social Psychology Bulletin, 36*, 443–454.
- McGuire, W. J. (1964). Inducing resistance to persuasion: Some contemporary approaches. In L. Berkowitz (ed.), *Advances in Experimental Social Psychology* (vol. 19, pp. 191–229). New York: Academic Press.
- Miller, R. C. (1976). Mere exposure, psychological reactance, and attitude change. *Public Opinion Quarterly, 40*, 229–233.
- Mills, J., & Jellison, J. M. (1967). Effect on opinion change of how desirable the communication is to the audience the communicator addressed. *Journal of Personality and Social Psychology, 6*, 98–101.
- Morrison, K. R. (2011). A license to speak up: Outgroup minorities and opinion expression. *Journal of Experimental Social Psychology, 47*, 756–766.
- Morrison, K. R., & Wheeler, S. C. (2010). Nonconformity defines the self: The role of minority opinion status in self-concept clarity. *Personality and Social Psychology Bulletin, 36*, 297–308.
- Moscovici, S. (1980). Toward a theory of conversion behavior. In L. Berkowitz (ed.), *Advances in Experimental Social Psychology* (vol. 13, pp. 209–239). New York: Academic Press.
- Moscovici, S. (1985). Social influence and conformity. In G. Lindzey & E. Aronson (eds.), *The Handbook of Social Psychology* (vol. 2, 3rd ed., pp. 347–412). New York: Random House.
- O’Neal, E. C., Kipnis, D., & Craig, K. M. (1994). Effects on the persuader of employing a peripheral route technique. *Basic and Applied Social Psychology, 15*, 225–238.
- Pechmann, C. (1992). Predicting when two-sided ads will be more effective than one-sided ads: The role of correlational and correspondent inferences. *Journal of Marketing Research, 29*, 441–453.
- Pelham, B. W., Sumarta, T. T., & Myaskovsky, L. (1994). The easy path from many to much: The numerosity heuristic. *Cognitive Psychology, 26*, 103–133.
- Perkins, A. W., & Forehand, M. R. (2010). Implicit social cognition and indirect measures in consumer behavior. In B. Gawronski & B. K. Payne (eds.),

- Handbook of Implicit Social Cognition: Measurement, Theory, and Applications* (pp. 535–547). New York: Guilford Press.
- Petrocelli, J. V., Tormala, Z. L., & Rucker, D. D. (2007). Unpacking attitude certainty: Attitude clarity and attitude correctness. *Journal of Personality and Social Psychology*, 92, 30–41.
- Petty, R. E. (1997). The evolution of theory and research in social psychology: From single to multiple effect and process models. In C. McGarty & S. A. Haslam (eds.), *The Message of Social Psychology: Perspectives on Mind in Society* (pp. 268–290). Oxford: Blackwell.
- Petty, R. E., & Briñol, P. (2008). Persuasion: From single to multiple to meta-cognitive processes. *Perspectives on Psychological Science*, 3, 137–147.
- Petty, R. E., & Briñol, P. (2012). The Elaboration Likelihood Model. In P. A. M. Van Lange, A. Kruglanski, & E. T. Higgins (eds.), *Handbook of Theories of Social Psychology* (vol. 1, pp. 224–245). London: Sage.
- Petty, R. E., & Briñol, P. (2015). Emotion and persuasion: Cognitive and meta-cognitive processes impact attitudes. *Cognition and Emotion*, 29, 1–26.
- Petty, R. E., Briñol, P., & DeMarree, K. G. (2007). The Meta-Cognitive Model (MCM) of attitudes: Implications for attitude measurement, change, and strength. *Social Cognition*, 25, 609–642.
- Petty, R. E., Briñol, P., Loersch, C., & McCaslin, M. J. (2009). The Need for cognition. In M. R. Leary & R. H. Hoyle (eds.), *Handbook of Individual Differences in Social Behavior* (pp. 318–329). New York: Guilford Press.
- Petty, R. E., Briñol, P. & Tormala, Z. L. (2002) Thought confidence as a determinant of persuasion: The self-validation hypothesis. *Journal of Personality and Social Psychology*, 82, 722–741.
- Petty, R. E., Briñol, P., Tormala, Z. L., & Wegener, D. T. (2007). The role of meta-cognition in social judgment. In E. T. Higgins & A. W. Kruglanski (eds.) *Social Psychology: A Handbook of Basic Principles* (2nd ed., pp. 254–284). New York: Guilford Press.
- Petty, R. E., & Cacioppo, J. T. (1979). Issue involvement can increase or decrease persuasion by enhancing message-relevant cognitive responses. *Journal of Personality and Social Psychology*, 37, 1915–1926.
- Petty, R. E., & Cacioppo, J. T. (1984). The effects of involvement on responses to argument quantity and quality: Central and peripheral routes to persuasion. *Journal of Personality and Social Psychology*, 46, 69–81.
- Petty, R. E., & Cacioppo, J. T. (1986). *Communication and Persuasion: Central and Peripheral Routes to Attitude Change*. New York: Springer-Verlag.
- Petty, R. E., Cacioppo, J. T. & Goldman, R. (1981). Personal involvement as a determinant of argument-based persuasion. *Journal of Personality and Social Psychology*, 41, 847–855.
- Petty, R. E., Cacioppo, J. T., & Schumann, D. (1983). Central and peripheral routes to advertising effectiveness: The moderating role of involvement. *Journal of Consumer Research*, 10, 135–146.
- Petty, R. E., & Fazio, R. H., & Briñol, P., (2009). The new implicit measures: An overview. In R. E. Petty, R. H. Fazio, & P. Briñol (eds.), *Attitudes: Insights from the New Implicit Measures* (pp. 3–18). New York: Psychology Press.
- Petty, R. E., Haugtvedt, C. P., & Smith, S. M. (1995). Elaboration as a determinant of attitude strength: Creating attitudes that are persistent, resistant, and predictive

- of behavior. In R. E. Petty & J. A. Krosnick (eds.), *Attitude Strength: Antecedents and Consequences*. (pp. 93–130). Mahwah, NJ: Erlbaum.
- Petty, R. E., & Krosnick, J. A. (1995) (eds.). *Attitude Strength: Antecedents and Consequences*. Hillsdale, NJ: Erlbaum.
- Petty, R. E., Schumann, D. W., Richman, S. A., & Strathman, A. J. (1993). Positive mood and persuasion: Different roles for affect under high and low elaboration conditions. *Journal of Personality and Social Psychology*, 64, 5–20.
- Petty, R. E., Tormala, Z. L., Briñol, P., & Jarvis, W. B. G. (2006). Implicit ambivalence from attitude change: An exploration of the PAST model. *Journal of Personality and Social Psychology*, 90, 21–41.
- Petty, R. E., Tormala, Z. L., & Rucker, D. D. (2004). Resisting persuasion by counter-arguing: An attitude strength perspective. In J. T. Jost, M. R. Banaji, & D. A. Prentice (eds.), *Perspectivism in Social Psychology: The Yin and Yang of Scientific Progress* (pp. 37–51). Washington, DC: American Psychological Association.
- Petty, R. E., & Wegener, D. T. (1998). Attitude change: Multiple roles for persuasion variables. In D. Gilbert, S. Fiske, & G. Lindzey (eds.), *The Handbook of Social Psychology* (4th ed., vol. 1, pp. 323–390). New York: McGraw-Hill.
- Petty, R. E., Wells, G. L., & Brock, T. C. (1976). Distraction can enhance or reduce yielding to propaganda: Thought disruption versus effort justification. *Journal of Personality and Social Psychology*, 34, 874–884.
- Pornpitakpan, C. (2004). The persuasiveness of source credibility: A critical review of five decades' evidence. *Journal of Applied Social Psychology*, 34, 243–281.
- Prislin, R., Boyle, S., Davenport, C., Farley, A., Jacobds, E., ichalak, J., Uehara, K., Zandian, F., & Xu, Y. (2011). On being influenced while trying to persuade: The feedback of persuasion outcomes to the persuader. *Social Psychological and Personality Science*, 2, 51–58.
- Rhine, R., & Severance, L. (1970). Ego-involvement, discrepancy, source credibility, and attitude change. *Journal of Personality and Social Psychology*, 16, 175–190.
- Rhodes, N., & Wood, W. (1992). Self-esteem and intelligence affect influenciability: The mediating role of message reception. *Psychological Bulletin*, 111, 156–171.
- Rind, B., & Kipnis, D. (2002). Changes in self-perceptions as a result of successfully persuading others. *Journal of Social Issues*, 55, 141–156.
- Rios, K., DeMarree, K. G., & Statzer, J. (2014). Attitude certainty and conflict style: Divergent effects of correctness and clarity. *Personality and Social Psychology Bulletin*, 40, 819–830.
- Rosenberg, M. J., & Hovland, C. I. (1960). Cognitive, affective, and behavioral components of attitudes. In C. I. Hovland, & M. J. Rosenberg (eds.), *Attitude Organization and Change: An Analysis of Consistency among Attitude Components*. New Haven, CT: Yale University Press
- Rucker, D. D., & Galinsky, A. D. (this volume). Power and consumer behavior. In M. I. Norton, D. D. Rucker, & C. Lamberton (eds.), *Cambridge Handbook of Consumer Psychology*. New York: Cambridge University Press.
- Rucker, D. D., & Petty, R. E. (2004). When resistance is futile: Consequences of failed counterarguing for attitude certainty. *Journal of Personality and Social Psychology*, 86, 219–235.

- Rucker, D. D., Petty, R. E., & Briñol, P. (2008). What's in a frame anyway? A meta-cognitive analysis of the impact of one versus two sided message framing on attitude certainty. *Journal of Consumer Psychology, 18*, 137–149.
- Rucker, D. D., & Tormala, Z. L. (2012). Meta-cognitive theory in consumer research. In P. Briñol & K. DeMarree (eds.), *Social Metacognition* (pp. 303–321). New York: Psychology Press.
- Rucker, D. D., Tormala, Z. L., Petty, R. E., & Briñol, P. (2014). Consumer conviction and commitment: An appraisal-based framework for attitude certainty. *Journal of Consumer Psychology, 24*, 119–36.
- Sanbonmatsu, D. M., Kardes, F. R., & Herr, P. M. (1992). The role of prior knowledge and missing information in multiattribute evaluation. *Organizational Behavior and Human Decision Processes, 51*(1), 76–91.
- Schwarz, N. (2004). Meta-cognitive experiences in consumer judgment and decision making. *Journal of Consumer Psychology, 14*, 332–348.
- Schwarz, N., Bless, H., & Bohner, G. (1991). Mood and persuasion: Affective status influence the processing of persuasive communications. In M. P. Zanna (ed.), *Advances in Experimental Social Psychology* (vol. 24, pp. 161–197). San Diego, CA: Academia Press.
- Schwarz, N., Bless, H., Strack, F., Klumpp, G., Rittenauer-Schatka, H., & Simons, A. (1991). Ease of retrieval as information: Another look at the availability heuristic. *Journal of Personality and Social Psychology, 61*, 195–202.
- See, Y. H. M., Fabrigar, L. R., & Petty, R. E. (2013). Affective-cognitive meta-bases versus structural bases of attitudes predict processing interest versus efficiency. *Personality and Social Psychology Bulletin, 39*, 1111–1123.
- See, Y. H. M., Petty, R. E., & Fabrigar, L. R. (2008). Affective and cognitive meta-bases of attitudes: Unique effects on information interest and persuasion. *Journal of Personality and Social Psychology, 94*, 938–955.
- Sela, A., & Berger, J. (2012). How attribute quantity influences option choice. *Journal of Marketing Research, 49*, 942–953.
- Shavitt, S., Lowrey, T., Han, S., (1992). Attitude functions in advertising: The interactive role of products and self-monitoring. *Journal of Consumer Psychology, 1*, 337–364.
- Sherif, M., & Hovland, C. I. (1961). *Social Judgment: Assimilation and Contrast Effects in Communication and Attitude Change*. New Haven, CT: Yale University Press.
- Shestowsky, D., Wegener, D. T., & Fabrigar, L. R. (1998). Need for cognition and interpersonal influence: Individual differences in impact on dyadic decisions. *Journal of Personality and Social Psychology, 74*, 1317–1328.
- Slovic, P., Finucane, M. L., Peters, E., & Macgregor, D. G. (2002). The Affect Heuristic. In T. Gilovich, D. Griffin, & D. Kahneman (eds.), *Heuristics and Biases: The Psychology of Intuitive Judgment* (pp. 397–420). New York: Cambridge University Press.
- Smith, C. T., De Houwer, J., & Nosek, B. A. (2013). Consider the source: Persuasion of implicit evaluations is moderated by manipulations of source credibility. *Personality and Social Psychology Bulletin, 39*, 193–205.
- Smith, S. M., Fabrigar, L. R., MacDougall, B. L., & Wiesenhal, N. L. (2008). The role of amount cognitive elaboration, and structural consistency of attitude-relevant knowledge in the formation of attitude certainty. *European Journal of Social Psychology, 38*, 280–295.

- Snyder, M., & Rothbart, M. (1971). Communicator attractiveness and opinion change. *Canadian Journal of Behavioural Science*, 3, 377–387.
- Staats, A. W., & Staats, C. (1958). Attitudes established by classical conditioning. *Journal of Abnormal and Social Psychology*, 67, 159–167.
- Tesser, A. (1978). Self-generated attitude change. In I. Berkowitz (ed.), *Advances in Experimental Social Psychology* (vol. 11, pp. 289–338). New York: Academic Press.
- Thompson, M. M., Zanna, M. P., & Griffin, D. W. (1995). Let's not be indifferent about (attitudinal) ambivalence. In R. E. Petty & J. A. Krosnick (eds.), *Attitude Strength: Antecedents and Consequences*. Hillsdale, NJ: Erlbaum.
- Tormala, Z. L. (2008). A new framework for resistance to persuasion: The resistance appraisals hypothesis. In W. D. Crano & R. Prislin (eds.), *Attitudes and Attitude Change* (pp. 213–234). New York: Psychology Press.
- Tormala, Z. L., Briñol, P., & Petty, R. E. (2006). When credibility attacks: The reverse impact of source credibility on persuasion. *Journal of Experimental Social Psychology*, 42, 684–691.
- Tormala, Z. L., Briñol, P., & Petty, R. E. (2007). Multiple roles for source credibility under high elaboration: It's all in the timing. *Social Cognition*, 25, 4, 536–552.
- Tormala, Z. L., Clarkson, J. J., & Henderson, M. D. (2011). Does fast or slow evaluation foster greater certainty? *Personality and Social Psychology Bulletin*, 37, 422–434.
- Tormala, Z. L., Clarkson, J. J., & Petty, R. E. (2006). Resisting persuasion by the skin of one's teeth: The hidden success of resisted persuasive messages. *Journal of Personality and Social Psychology*, 91 (3), 423–435.
- Tormala, Z. L., DeSensi, V. L., & Petty, R. E. (2007). Resisting persuasion by illegitimate means: A meta-cognitive perspective on minority influence. *Personality and Social Psychology Bulletin*, 33, 354–367.
- Tormala, Z. L., Falces, C., Briñol, P., & Petty, R. E. (2007). Ease of retrieval effects in social judgment: The role of unrequested cognitions. *Journal of Personality and Social Psychology*, 93, 143–157.
- Tormala, Z. L., & Petty, R. E. (2002). What doesn't kill me makes me stronger: The effects of resisting persuasion on attitude certainty. *Journal of Personality and Social Psychology*, 83, 1298–1313.
- Tormala, Z. L., & Petty, R. E. (2004a). Source credibility and attitude certainty: A metacognitive analysis of resistance to persuasion. *Journal of Consumer Psychology*, 14, 427–442.
- Tormala, Z. L., & Petty, R. E. (2004b). Resistance to persuasion and attitude certainty: The moderating role of elaboration. *Personality and Social Psychology Bulletin*, 30, 1446–1457.
- Tormala, Z. L., Petty, R. E., & Briñol, P. (2002). Ease of retrieval effects in persuasion: A self-validation analysis. *Personality and Social Psychology Bulletin*, 28, 1700–1712.
- Tormala, Z. L., Petty, R. E., & DeSensi, V. L. (2010). Multiple roles for minority sources in persuasion and resistance. In R. Martin & M. Hewstone (eds.), *Minority Influence and Innovation: Antecedents, Processes, and Consequences* (pp. 105–131). London: Psychology Press.
- Tormala, Z. L., & Rucker, D. D. (2007). Attitude certainty: A review of past findings and emerging perspectives. *Social and Personality Psychology Compass*, 1, 469–492.

- Visser, P. S., & Mirabile, R. R. (2004). Attitudes in the social context: The impact of social network composition on individual-level attitude strength. *Journal of Personality and Social Psychology*, 87, 779–795.
- Wan, E. W., Rucker, D. D., Tormala, Z. L., & Clarkson, J. J. (2010). The effect of regulatory depletion on attitude certainty. *Journal of Marketing Research*, 47, 531–541.
- Wänke, M., Bohner, G., & Jurkowitsch, A. (1997). There are many reasons to drive a BMW: Does imagined ease of argument generation influence attitudes? *Journal of Consumer Research*, 24, 170–177.
- Wells, G. L., & Petty, R. E. (1980). The effects of overt head movements on persuasion: Compatibility and incompatibility of responses. *Basic and Applied Social Psychology*, 1, 219–230.
- Wood, W., Lundgren, S., Quелlette, J. A., Busceme, S., & Blackstone, T. (1994). Minority influence: A meta-analytic review of social influence processes. *Psychological Bulletin*, 115, 323–345.
- Xu, A. J., & Wyer, R. S. (2012). The role of bolstering and counterarguing mindsets in persuasion. *Journal of Consumer Research*, 38, 920–932.
- Zajonc, R. B. (1968). Attitudinal effects of mere exposure. *Journal of Personality and Social Psychology*, 9, 1–27.
- Zanna, M. P., & Rempel, J. K. (1988). Attitudes: A new look at an old concept. In D. Bar-Tal & A. W. Kruglanski (eds.), *The Social Psychology of Attitudes* (pp. 315–334). New York: Cambridge University Press.

3 Consumer Prediction

Forecasted Utility, Psychological Distance, and Their Intersection

Carey K. Morewedge and Hal E. Hershfield

Consumer prediction encompasses the cognitive, affective, and motivational psychological processes by which consumers anticipate (and subsequently produce) the future. Prediction is a pervasive factor in consumer decision making, from everyday decisions such as which lunch one should purchase to major decisions about how much one will need to save for retirement. More generally, predictions are the method by which consumers determine which choice options will bring them the greatest satisfaction in the present and by which they anticipate their needs and wants in the near and distant future. In this chapter, we examine the processes by which consumers infer whether and what will happen in the future, the accuracy of their predictions, their antecedents, and consequents. We (i) review the two dimensions of prediction that have been most studied, utility and psychological distance, and how they combine to determine the perceived value of prospects (choice options). We (ii) discuss innovative research on these topics over the last decade, and (iii) end with important open questions and promising future directions.

I. Dimensions of Prediction

Guided by the evaluation of prospective (future) events in terms of their expected value (Kahneman & Tversky, 1979; Von Neumann & Morgenstern, 1947), the two dimensions of prediction typically studied by consumer psychologists are (a) the utility of future events and (b) their probability of occurrence. In this chapter, we discuss these two dimensions of prediction in terms of their recently refined and expanded definitions. Specifically, we review the concept of utility in terms of a refined definition that distinguishes between indirect and direct measures of utility. We review the concept of probability as an instance of a broader dimension of psychological distance, which also includes time, physical space, and more abstract forms of distance such as social connection.

Utility

Utility is a measure of the value of a stimulus that typically connotes the total pleasure or pain associated with its anticipation, experience, and recollection.